TECH WITH SHIVANSH

STATISTICS INTERVIEW PREPARATION QUESTION BANK

LEVEL 1 — FOUNDATIONS

60 Industry-Oriented Questions | Solved Numericals | Manual & Python Practice
Data Analytics • Data Science • Foundational Machine Learning

How to Use This Question Bank

- Attempt each question before reading the answer. Explain the reasoning aloud as you would in an interview.
- For numericals, write the formula, substitute values, calculate the result, and state what it means in business terms.
- For Python questions, understand the calculation rather than memorizing code. Check whether a function uses sample or population conventions.
- For scenario questions, identify the business objective, data type, suitable method, assumptions, limitations, and recommendation.
- Company-specific attribution is included only where public sources support it. Most questions are tagged as common interview-preparation themes or original high-probability practice questions; they are not falsely attributed to a named employer.

Level 1 Coverage Map

Research and Attribution Policy

The question themes were cross-checked against public interview-preparation resources including DataCamp's statistics and data-science interview guides, Exponent's statistics interview guide, a public Naukri Campus statistics interview PDF, and publicly shared interview-question collections. These sources support topic selection and common interview themes, but they do not establish that every question below was asked by a specific employer. Questions are therefore labelled as common themes or practice questions unless a source explicitly documents an attribution.

- DataCamp — Statistics Interview Questions: https://www.datacamp.com/blog/statistics-interview-questions
- DataCamp — Data Scientist Interview Questions: https://www.datacamp.com/blog/data-scientist-interview-questions
- Exponent — Statistics Data Science Interview Questions: https://www.tryexponent.com/blog/top-statistics-data-science-interview-questions
- Naukri Campus — Statistics in Data Science Interview Questions (PDF): https://www.naukri.com/campus/career-guidance/wp-content/uploads/2024/01/Statistics-in-Data-Science-Interview-Questions.pdf
- Public question collection — Statistics Interview Questions for Data Scientists: https://github.com/youssefHosni/Data-Science-Interview-Questions-Answers

Note: Public interview reports and prep guides are not a guarantee of what a particular employer will ask. Always check role-specific requirements.

SECTION A — FOUNDATIONAL CONCEPTS

Q1. What is statistics, and why is it important in data analytics? [★ ☆ ☆]

ANSWER

Statistics is the discipline of collecting, organizing, summarizing, analysing, and interpreting data under uncertainty. In analytics, it helps describe performance, compare groups, estimate unknown quantities, test claims, and support decisions. Example: estimating whether a new checkout page improves conversion rather than judging it from a few anecdotes.

Follow-up question: Give one example where a summary statistic could mislead a decision.

Answer: A high average delivery time may be driven by a few extreme delays and may not represent the typical customer experience.

Q2. Differentiate descriptive and inferential statistics. [★ ☆ ☆]

ANSWER

Descriptive statistics summarize the data observed, such as average order value in this month's transactions. Inferential statistics use sample data to estimate or test claims about a wider population, with uncertainty quantified through methods such as confidence intervals and hypothesis tests.

Q3. Differentiate population, sample, parameter, and statistic. [★ ☆ ☆]

ANSWER

A population is the complete group of interest; a sample is a subset. A parameter describes the population, such as population mean μ. A statistic is calculated from a sample, such as sample mean $\bar{x}$, and may be used to estimate a parameter.

Q4. Census vs. sampling: when would you use each? [★ ☆ ☆]

ANSWER

A census measures every unit in the defined population; sampling measures a subset. A census can be costly or slow and may still contain measurement errors. Sampling is faster and cheaper but introduces sampling uncertainty and possible selection bias.

Q5. What is the difference between qualitative and quantitative data? [★ ☆ ☆]

ANSWER

Qualitative data represents categories or attributes, such as payment method. Quantitative data represents numerical amounts or counts, such as order value or number of orders. Numeric labels for categories do not automatically make the data quantitative.

Follow-up question: Is a customer ID a quantitative variable?

Answer: No. It is an identifier used to label customers; arithmetic operations such as calculating its mean have no meaningful interpretation.

Q6. Differentiate discrete and continuous variables. [★ ☆ ☆]

ANSWER

A discrete variable takes countable values, often whole-number counts, such as number of returns. A continuous variable can take any value in a range at the measurement precision used, such as delivery time or package weight.

Q7. Explain nominal, ordinal, interval, and ratio measurement scales. [★ ☆ ☆]

ANSWER

Nominal categories have no inherent order; ordinal categories have order but unequal spacing is not guaranteed. Interval scales have meaningful differences but no true zero; ratio scales have meaningful differences and a true zero. Examples: region, satisfaction rating, temperature in °C, and revenue.

Q8. What is sampling bias? Give a business example and a remedy. [★ ☆ ☆]

ANSWER

Sampling bias occurs when the selection process systematically over- or under-represents parts of the target population. Example: surveying only app users to measure satisfaction among all customers, including people who shop in stores. Define the target population, use an appropriate sampling frame, and consider random or stratified sampling.

Q9. Compare simple random, systematic, stratified, and cluster sampling. [★ ★ ★]

ANSWER

Simple random sampling selects units randomly from the population. Systematic sampling selects every kth unit after a random start. Stratified sampling samples within meaningful subgroups. Cluster sampling randomly selects groups or clusters and samples units within selected clusters. Each method depends on the sampling frame and design.

Follow-up question: A retailer wants adequate representation of rural and urban customers. Which method may help?

Answer: Stratified sampling: divide customers into rural and urban groups, then sample from each group.

Q10. Differentiate sampling error and non-sampling error. [★ ☆ ☆]

ANSWER

Sampling error arises because a sample rather than the entire population is measured; it can often be quantified under a valid design. Non-sampling errors include measurement mistakes, missing responses, data-entry errors, and coverage problems; a larger sample does not necessarily eliminate them.

Q11. Primary vs. secondary data: what are the trade-offs? [★ ☆ ☆]

ANSWER

Primary data is collected directly for the current objective, offering control over questions and measurements but often requiring more time and cost. Secondary data already exists, such as CRM exports or public datasets, and is faster to obtain but may not match the exact question or definitions.

Q12. What is the difference between a variable and an observation? [★ ☆ ☆]

ANSWER

A variable is a characteristic recorded, such as order value. An observation is one recorded unit or case, often a row in a dataset. In a customer table, a row may be one customer and columns may be age band, region, and lifetime spend; the row unit must be clearly defined.

SECTION B — DESCRIPTIVE STATISTICS AND NUMERICALS

Q13. Calculate the mean of 12, 15, 18, 20, and 25. [★★☆]

SOLVED NUMERICAL

Formula: Mean = sum of observations ÷ number of observations

Substitute: Mean = (12 + 15 + 18 + 20 + 25) ÷ 5

Calculate: Mean = 90 ÷ 5 = 18

Interpretation: The average observation is 18. The mean uses every value and is sensitive to extreme observations.

Follow-up question: If 25 were actually 125, what would happen to the mean?

Answer: The total increases by 100, so the mean increases by 100 ÷ 5 = 20, from 18 to 38.

Q14. For salaries ₹25,000, ₹28,000, ₹30,000, ₹32,000, and ₹150,000, should you report mean or median as the typical salary? [★★☆]

SOLVED NUMERICAL

Sorted salaries: ₹25,000, ₹28,000, ₹30,000, ₹32,000, ₹150,000

Median: The middle (3rd) value = ₹30,000

Mean formula: Sum of salaries ÷ number of salaries

Substitute: (25,000 + 28,000 + 30,000 + 32,000 + 150,000) ÷ 5

Calculate: ₹265,000 ÷ 5 = ₹53,000

Interpretation: The high salary pulls the mean upward; the median better represents the middle employee in this small set.

Q15. What is a weighted mean? Solve: scores 70, 80, and 90 have weights 20%, 30%, and 50%. [★★☆]

SOLVED NUMERICAL

Formula: Weighted mean = Σ(value × weight)

Substitute: 70(0.20) + 80(0.30) + 90(0.50)

Calculate: 14 + 24 + 45 = 83

Check: 0.20 + 0.30 + 0.50 = 1.00

Interpretation: The weighted average is 83 because the three components contribute unequally.

Q16. Find the median of 4, 7, 9, 12, 15, 18. [★★☆]

SOLVED NUMERICAL

Step 1: There are 6 observations, an even number.

Step 2: Identify the two middle values: 9 and 12 (3rd and 4th).

Formula: Median = (lower middle value + upper middle value) ÷ 2

Calculate: (9 + 12) ÷ 2 = 21 ÷ 2 = 10.5

Interpretation: The median is 10.5.

Q17. What is the mode, and when is it useful? [★☆☆]

ANSWER

The mode is the most frequent value or category. It is useful for categorical data and identifying the most common item, such as the most-used payment method. A dataset may have no unique mode or multiple modes.

Follow-up question: Can a categorical variable such as product category have a mode? Yes.

Answer: Yes. The mode is the most frequently occurring category.

Q18. Calculate the range of 14, 19, 22, 28, and 41. [★★☆]

SOLVED NUMERICAL

Formula: Range = maximum − minimum

Substitute: Range = 41 − 14

Calculate: Range = 27

Interpretation: The observations span 27 units. The range depends only on the two extreme values.

Q19. Differentiate population variance and sample variance. [★☆☆]

ANSWER

Population variance divides the squared deviations from the population mean by N. Sample variance typically divides by n − 1 to correct the downward bias of the sample variance estimator under independent sampling. Use the convention matching the question and software function.

Q20. Calculate population variance and standard deviation for 2, 4, 6. [★★★]

SOLVED NUMERICAL

Data: 2, 4, 6

Step 1 — Mean: (2 + 4 + 6) ÷ 3 = 12 ÷ 3 = 4

Step 2 — Squared deviations: $(2-4)^2 = 4$; $(4-4)^2 = 0$; $(6-4)^2 = 4$

Step 3 — Sum of squared deviations: 4 + 0 + 4 = 8

Population variance: 8 ÷ 3 ≈ 2.667

Population SD: √2.667 ≈ 1.633

If these are a sample: sample variance = 8 ÷ (3−1) = 4; sample SD = √4 = 2.

Q21. What does standard deviation tell you? [★☆☆]

ANSWER

Standard deviation measures spread around the mean in the original units. A low standard deviation indicates values are relatively close to the mean; a high value indicates more variability. It is sensitive to outliers and does not by itself describe the distribution shape.

Follow-up question: Can two datasets have the same mean but different standard deviations? Yes.

Answer: Yes. For example, [4, 5, 6] and [0, 5, 10] both have mean 5 but different spreads.

Q22. Calculate the coefficient of variation for mean = 50 and standard deviation = 5. [★★☆]

SOLVED NUMERICAL

Formula: CV = (standard deviation ÷ mean) × 100%

Substitute: CV = (5 ÷ 50) × 100%

Calculate: 0.10 × 100% = 10%

Interpretation: Standard deviation is 10% of the mean. CV is most meaningful when the mean is positive and not near zero.

Q23. What is mean absolute deviation (MAD), and how does it differ from standard deviation? [★☆☆]

ANSWER

Mean absolute deviation is the average absolute distance from a chosen centre, commonly the mean or median

depending on convention. Standard deviation uses squared deviations and then takes a square root, giving relatively greater influence to large deviations. Always specify which MAD convention is being used.

Q24. What are quartiles, percentiles, and the five-number summary? [★☆☆]

ANSWER

Percentiles divide ordered data by rank; the 25th, 50th, and 75th percentiles correspond conceptually to Q1, median, and Q3. The five-number summary is minimum, Q1, median, Q3, and maximum. Exact quartile results can differ across software because quantile algorithms vary.

Q25. Dataset A has mean 100 and SD 10; dataset B has mean 20 and SD 4. Which has greater relative variability? [★☆☆]

ANSWER

CV(A) = 10/100 × 100% = 10%. CV(B) = 4/20 × 100% = 20%. Dataset B has greater relative variability. This comparison is useful only if both means are positive and the CV is appropriate for the data scale.

Follow-up question: Which has the larger absolute SD? Dataset A.

Answer: Dataset A has the larger standard deviation, meaning its observations are more spread out around the mean in absolute units.

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SECTION C — DISTRIBUTIONS, OUTLIERS, AND STANDARDIZATION

Q26. What is a positively skewed distribution? [★ ☆ ☆]

ANSWER

A positively or right-skewed distribution has a longer tail toward higher values. A few large observations may pull the mean above the median, though the exact ordering depends on the distribution. Example: customer spending with a small number of very large purchases.

Q27. What is the difference between a histogram and a bar chart? [★ ☆ ☆]

ANSWER

A histogram shows the distribution of numerical data grouped into bins; adjacent bins typically touch because the intervals are ordered and continuous. A bar chart compares categories, and gaps usually separate the bars. Choosing bins can change a histogram's appearance.

Q28. How do you identify potential outliers using the IQR rule? [★ ☆ ☆]

ANSWER

Calculate IQR = Q3 − Q1. The common 1.5×IQR fences are Q1 − 1.5×IQR and Q3 + 1.5×IQR. Values beyond these fences are flagged as potential outliers, not automatically errors or values that must be removed.

Q29. Solve the IQR outlier problem: Q1 = 12, Q3 = 20. Find the fences. [★ ★ ★]

SOLVED NUMERICAL

Formula: IQR = Q3 − Q1

Substitute: IQR = 20 − 12 = 8

Lower fence: Q1 − 1.5 × IQR = 12 − 1.5(8) = 12 − 12 = 0

Upper fence: Q3 + 1.5 × IQR = 20 + 1.5(8) = 20 + 12 = 32

Interpretation: Values below 0 or above 32 are flagged as potential outliers, not automatically errors.

Follow-up question: Does a value above 32 prove a data-entry error? No.

Answer: No. The IQR rule flags a potential outlier; investigate the source and context before deciding whether it is an error.

Q30. What is a Z-score? Calculate it for x = 70, mean = 50, SD = 10. [★ ★ ☆]

SOLVED NUMERICAL

Formula: Z = (observation − mean) ÷ standard deviation

Substitute: Z = (70 − 50) ÷ 10

Calculate: Z = 20 ÷ 10 = 2

Interpretation: The observation is 2 standard deviations above the mean. The Z-score alone does not prove it is an error or outlier.

Q31. Explain the empirical rule for a normal distribution. [★ ★ ☆]

SOLVED NUMERICAL

Empirical rule (approximately normal distribution):

Within 1 SD: about 68%

Within 2 SDs: about 95%

Within 3 SDs: about 99.7%

Interpretation: These are approximate proportions and should not be assumed for strongly skewed or otherwise non-

normal distributions.

Q32. What does it mean to standardize data? [★ ☆ ☆]

ANSWER

A common standardization transforms x to z = (x − mean) / SD, placing values in standard-deviation units. Standardization is often useful when features have different scales, but it is not the same as min-max normalization and can be affected by outliers.

Q33. Differentiate normalization and standardization. [★ ☆ ☆]

ANSWER

Standardization typically centres values at the mean and scales by standard deviation. Min-max normalization rescales values to a specified range, often 0 to 1. The appropriate method depends on the model, feature distributions, and sensitivity to outliers.

Follow-up question: Should preprocessing parameters be fitted on the full dataset before a train/test split? No; fit them on training data to avoid leakage.

Answer: No. Fit preprocessing steps on training data only, then apply those learned transformations to the test data to avoid data leakage.

Q34. What does a box plot communicate? [★ ☆ ☆]

ANSWER

A box plot typically shows the median, quartiles, and whiskers based on a stated rule; points beyond whiskers may be shown individually. It is useful for comparing distributions across groups but hides detailed shape and multimodality.

Q35. Business scenario: delivery times are usually 20–40 minutes, but a few take 3 hours. Which summaries and visuals would you use? [★ ★ ☆]

ANSWER

Report median and IQR alongside mean and standard deviation, and inspect a histogram or box plot. Validate the 3-hour observations and investigate causes such as failed deliveries or system delays. Do not remove genuine delays simply because they are extreme; they may be operationally important.

SECTION D — INTRODUCTORY PROBABILITY

Q36. What is probability? Explain sample space and event. [★ ☆ ☆]

ANSWER

Probability quantifies uncertainty from 0 (impossible) to 1 (certain). A sample space is the set of possible outcomes; an event is a subset of those outcomes. For a fair six-sided die, the sample space is {1,2,3,4,5,6}, and the event "even number" is {2,4,6}.

Q37. What is the complement rule? [★ ☆ ☆]

ANSWER

For an event A, P(not A) = 1 − P(A). It is especially useful when "at least one" is easier to calculate through the probability of none.

Follow-up question: If the probability a package is delayed is 0.08, what is the probability it is not delayed? 0.92.

Answer: P(not delayed) = 1 − 0.08 = 0.92, or 92%.

Q38. A fair die is rolled twice. What is the probability of getting at least one six? [★ ★ ★]

SOLVED NUMERICAL

Step 1: P(no six on one roll) = 5/6

Step 2: For two independent rolls, P(no six on either roll) = (5/6) × (5/6) = 25/36

Step 3 — Complement: P(at least one six) = 1 − 25/36 = 11/36

Calculate: 11 ÷ 36 ≈ 0.3056 = 30.56%

Interpretation: The chance of at least one six in two rolls is about 30.56%.

Q39. Differentiate mutually exclusive and independent events. [★ ☆ ☆]

ANSWER

Mutually exclusive events cannot occur together, so $P(A \cap B)=0$. Independent events satisfy $P(A \cap B)=P(A)P(B)$, meaning occurrence of one does not change the probability of the other. Two events with positive probabilities cannot be both mutually exclusive and independent.

Q40. A bag contains 3 red and 2 blue balls. One ball is drawn at random. What is P(red)? [★ ★ ☆]

SOLVED NUMERICAL

Formula: P(event) = favourable outcomes ÷ total equally likely outcomes

Substitute: P(red) = 3 ÷ 5

Calculate: P(red) = 0.60 = 60%

Interpretation: The probability of drawing a red ball is 60%, assuming each ball is equally likely.

Q41. What is conditional probability? [★ ☆ ☆]

ANSWER

Conditional probability is the probability of event A given that event B has occurred: $P(A|B)=P(A \cap B)/P(B)$, provided $P(B)>0$. The condition changes the relevant information and may change the probability.

Follow-up question: Give a business example. Probability of purchase given a customer clicked an advertisement.

Q42. In a group, 40% use a mobile app, and 30% both use the app and purchase. Find P(purchase | app user). [★★★]

SOLVED NUMERICAL

Formula: P(purchase | app) = P(purchase and app) ÷ P(app)

Substitute: 0.30 ÷ 0.40

Calculate: 0.75 = 75%

Interpretation: Based on the supplied proportions, 75% of app users purchased. This conditional association does not by itself establish causation.

Q43. Explain Bayes' theorem in plain language. [★☆☆]

ANSWER

Bayes' theorem updates the probability of a hypothesis after observing evidence: $P(A|B)=P(B|A)P(A)/P(B)$. It combines the prior probability, how likely the evidence is under the hypothesis, and the overall probability of the evidence.

Q44. A disease affects 1% of people. A test has 90% sensitivity and a 5% false-positive rate. If a person tests positive, estimate the probability they have the disease. [★★★]

SOLVED NUMERICAL

Assume 10,000 people.

Step 1 — Disease cases: 1% × 10,000 = 100

Step 2 — True positives: 90% × 100 = 90

Step 3 — People without disease: 10,000 − 100 = 9,900

Step 4 — False positives: 5% × 9,900 = 495

Step 5 — All positive tests: 90 + 495 = 585

Formula: P(disease | positive) = true positives ÷ all positive tests

Calculate: 90 ÷ 585 ≈ 0.1538 = 15.4%

Interpretation: Even with 90% sensitivity, the positive predictive value is about 15.4% because the disease is rare and false positives arise from a much larger group.

Q45. A fair coin is tossed three times. What is the probability of exactly two heads? [★★★]

SOLVED NUMERICAL

Formula: P(exactly 2 heads) = number of valid sequences × probability of each sequence

Step 1: Number of ways = C(3,2) = 3

Step 2: Probability of each 3-toss sequence = $(1/2)^3$ = 1/8

Step 3: Multiply: 3 × 1/8 = 3/8

Calculate: 3 ÷ 8 = 0.375 = 37.5%

Interpretation: The probability of exactly two heads in three fair tosses is 37.5%.

Follow-up question: What assumptions support the binomial model? Fixed number of trials, independent trials, two outcomes per trial, constant success probability.

SECTION E — BUSINESS SCENARIOS AND METHOD SELECTION

Q46. Scenario: an online retailer wants to report "typical" order value, but a few corporate orders are extremely large. What would you report? [★★☆]

ANSWER

Report median and IQR to describe a typical order robustly, and optionally mean and standard deviation to quantify overall revenue-weighted behaviour. Show the distribution and explain the extreme orders. The correct choice depends on whether the question is about a typical order or total commercial value.

Q47. Scenario: customer satisfaction is recorded as Very dissatisfied, Dissatisfied, Neutral, Satisfied, Very satisfied. What data type is this, and what summary is suitable? [★★☆]

ANSWER

This is ordinal categorical data because the categories have a natural order, but equal spacing between levels is not guaranteed. Report category frequencies and percentages; the median category can also be useful. A mean score may be used only with an explicit scoring convention and caveat.

Q48. Scenario: a survey response rate is 8%, and only highly engaged customers tend to respond. What is the risk? [★★☆]

ANSWER

Nonresponse bias may occur if respondents differ systematically from nonrespondents on the outcome of interest. Compare available characteristics of respondents and nonrespondents, follow up with a subset where possible, and consider weighting or redesign. Weighting cannot guarantee removal of bias if key differences are unobserved.

Q49. Scenario: two stores have the same average daily sales but very different standard deviations. How should management interpret this? [★★☆]

ANSWER

Their average performance is similar, but sales consistency differs. Examine standard deviation, coefficient of variation where appropriate, distribution plots, seasonality, and unusual days. Higher variation may imply less predictable staffing or inventory needs.

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Follow-up question: What else should be checked before comparing variability? Store size, trading days, product mix, and comparable periods.

Q50. Scenario: a manager says, "Our sales increased after the advertising campaign, so the campaign caused the increase." How would you respond? [★★★]

ANSWER

The timing is suggestive but does not establish causation. Other factors—seasonality, pricing, competitor actions, or customer mix—may explain the increase. Use a randomized experiment when feasible, or a suitable quasi-experimental design and sensitivity checks when randomization is not possible.

Q51. Scenario: a dataset contains customer_id, age, annual_spend, and membership_tier. Identify the likely measurement types. [★★☆]

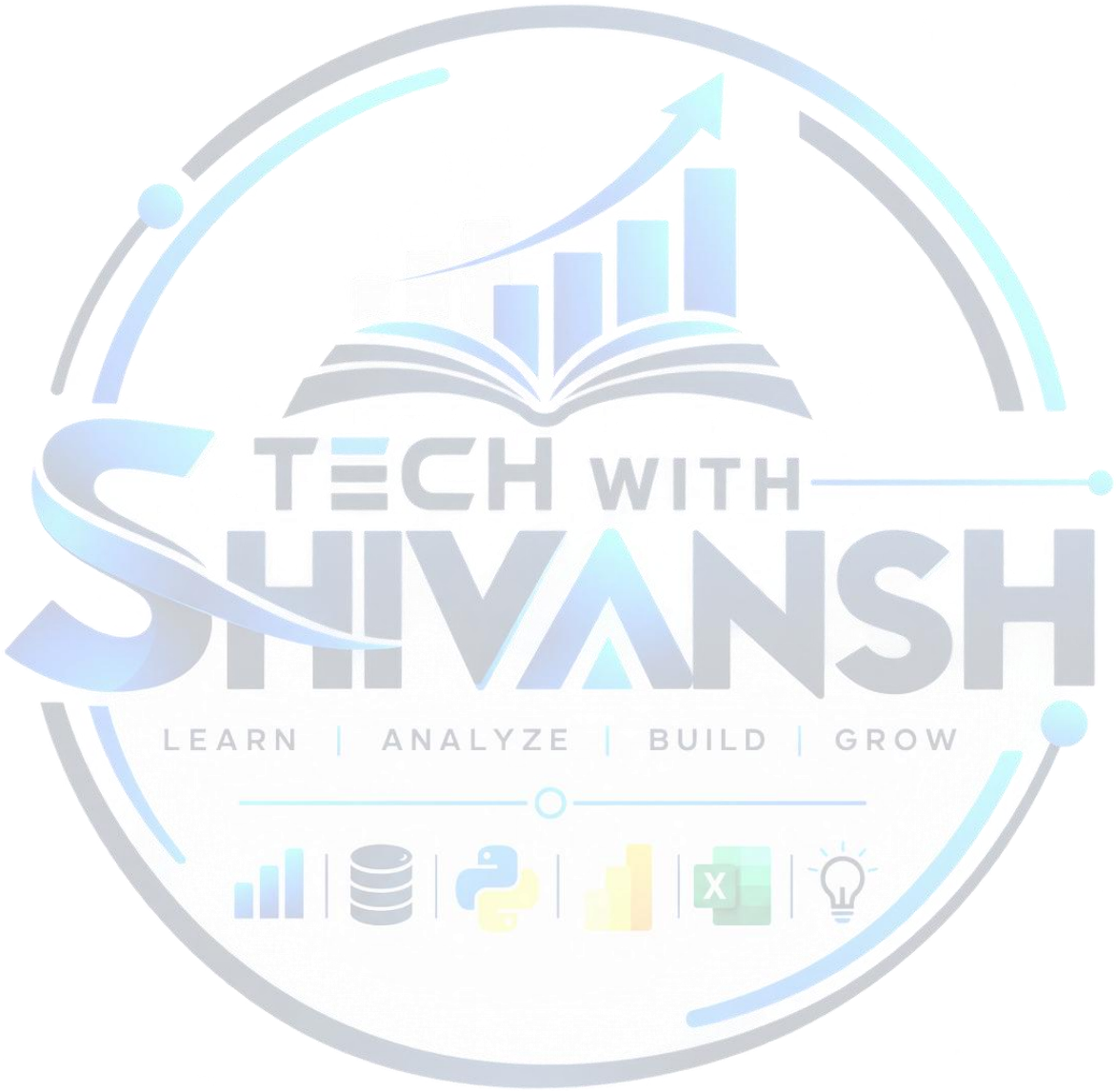
ANSWER

customer_id is an identifier, not a measured quantity; age and annual_spend are quantitative variables; membership_tier is ordinal categorical if its levels have a defined ranking. Confirm how age is stored and whether tier ordering is meaningful.

Q52. An analyst wants to compare average spending for two independent customer groups. What information is needed before choosing a test? [★★★]

ANSWER

Clarify the outcome scale, whether groups are independent or paired, sample sizes, distribution and outlier behaviour, variance assumptions, sampling design, and the target estimand. A two-sample t-test or Welch's t-test may be suitable for means; other methods may be preferable for strongly non-normal data or different estimands.



SECTION F — PYTHON, INTERPRETATION, AND INTERVIEW TRAPS

Q53. In NumPy, how do you calculate mean, median, population standard deviation, and sample standard deviation? [★☆☆]

PYTHON

```
import numpy as np
x = np.array([2, 4, 6])

print(np.mean(x))              # 4.0
print(np.median(x))            # 4.0
print(np.std(x, ddof=0))       # 1.633... population SD
print(np.std(x, ddof=1))       # 2.0 sample SD
```

Explanation

NumPy's ddof argument controls the divisor. ddof=0 uses N for population SD; ddof=1 uses n−1 for sample SD.

Follow-up question: What common mistake causes NumPy and pandas SD results to differ? Using different ddof defaults.

Q54. Write Python to calculate descriptive statistics for a pandas Series. [★☆☆]

PYTHON

```
import pandas as pd
s = pd.Series([12, 15, 18, 20, 25])

print(s.mean())
print(s.median())
print(s.std())
print(s.var())
print(s.describe())
```

Explanation

pandas Series.std() and var() use sample conventions by default (ddof=1). Check missing values and data types before interpreting the summary.

Q55. Python exercise: calculate a Z-score for x = 70 when mean = 50 and SD = 10. [★☆☆]

PYTHON

```
x = 70
mean = 50
sd = 10
z = (x - mean) / sd
print(z)  # 2.0
```

Explanation

The Z-score is 2, meaning the observation is two standard deviations above the mean.

Q56. A Python function returns np.std(data) = 1.63 for [2, 4, 6], while your sample SD calculation gives 2. Explain. [★★★]

PYTHON

```
import numpy as np
data = np.array([2, 4, 6])
```

```
print(np.std(data, ddof=0))  # population SD: about 1.633
print(np.std(data, ddof=1))  # sample SD: 2.0
```

Explanation
NumPy defaults to ddof=0. Use ddof=1 when calculating the usual sample standard deviation.

Q57. What checks should you perform before trusting descriptive statistics from a dataset? [★★☆]

ANSWER
Check the unit of analysis, missingness, duplicates, data types, impossible values, date ranges, units, and whether the sample represents the target population. Validate outliers against source records and compare totals with trusted business systems. Document exclusions and transformations.

Follow-up question: Why inspect the data before computing averages? Bad records can create plausible-looking but misleading summaries.

Q58. A candidate says, "A standard deviation of zero means the mean is zero." Is this correct? [★☆☆]

ANSWER
No. Standard deviation zero means every observed value is identical, so all values equal the mean; that common value may be any number. For example, [5,5,5] has mean 5 and SD 0.

Q59. A candidate says, "If the sample size doubles, standard error is cut in half." Is this correct for a sample mean? [★★★]

ANSWER
Not generally. SE = s/√n. If s remains approximately constant and sample size doubles, the new SE is the old SE/√2 ≈ 0.707 times the old SE, a reduction of about 29.3%, not 50%. To halve SE, sample size must increase by about four times.

Q60. Final interview scenario: you receive a customer dataset and are asked to describe customer spending in five minutes. What is your structured approach? [★★★]

ANSWER
First clarify the unit and target population. Check data types, missingness, duplicates, impossible values, and time period. Summarize count, mean, median, SD, quartiles, and IQR; inspect a histogram and box plot; compare relevant segments only when useful. Investigate extreme values, distinguish descriptive findings from causal claims, and finish with two or three decision-relevant insights and limitations.

Follow-up question: What should you avoid? Reporting only the mean, silently deleting outliers, or claiming causation from descriptive summaries.

Manual and Python Cross-Check Examples

Example 1 — Mean and Standard Deviation

Data: 2, 4, 6.

Mean = (2+4+6)/3 = 4.

Squared deviations total 8.

Population variance = 8/3;

population SD ≈ 1.633.

Sample variance = 8/(3−1) = 4; sample SD = 2.

PYTHON

```
import numpy as np;
x=np.array([2,4,6]);
np.mean(x) -> 4.0;
np.std(x, ddof=0) -> about 1.633;
np.std(x, ddof=1) -> 2.0.
```

Example 2 — Conditional Probability

Given P(App)=0.40 and P(Purchase ∩ App)=0.30,

P(Purchase | App)=0.30/0.40=0.75.

The interpretation is that 75% of app users purchased in the stated data; it is not proof that app use caused the purchase.

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Example 3 — Bayes / Base-Rate Reasoning

For 10,000 people, 1% prevalence means 100 affected. At 90% sensitivity, 90 true positives. Among 9,900 unaffected people, a 5% false-positive rate yields 495 false positives. Positive predictive value = 90/(90+495) ≈ 15.4%.

Python is useful for verification, but candidates should be able to explain the denominator and assumptions manually.

Level 1 Rapid-Revision Formula Sheet

Measure	Formula
Arithmetic mean	$\bar{x} = \Sigma x / n$
Weighted mean	$\Sigma(w_i x_i) / \Sigma w_i$
Range	Maximum − Minimum
Population variance	$\sigma^2 = \Sigma(x_i-\mu)^2 / N$
Sample variance	$s^2 = \Sigma(x_i-\bar{x})^2 / (n-1)$
Standard deviation	Square root of variance
Coefficient of variation	(SD / Mean) × 100%, when meaningful
Interquartile range	IQR = Q3 − Q1
IQR lower fence	Q1 − 1.5 × IQR
IQR upper fence	Q3 + 1.5 × IQR
Z-score	z = (x − mean) / SD
Complement probability	$P(A^c) = 1 - P(A)$
Conditional probability	$P(A\|B) = P(A \cap B) / P(B), P(B)>0$
Bayes theorem	$P(A\|B) = P(B\|A)P(A) / P(B), P(B)>0$

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Research References and Scope Notes

The references below informed the selection of recurring interview themes. They are preparation resources, not proof that each question was asked at a named company.

- DataCamp, “The Top 35 Statistics Interview Questions and Answers”: https://www.datacamp.com/blog/statistics-interview-questions
- DataCamp, “29 Top Data Scientist Interview Questions For All Levels”: https://www.datacamp.com/blog/data-scientist-interview-questions
- Exponent, “Top 25 Statistics Data Science Interview Questions”: https://www.tryexponent.com/blog/top-statistics-data-science-interview-questions
- Naukri Campus, “Statistics in Data Science Interview Questions”: https://www.naukri.com/campus/career-guidance/wp-content/uploads/2024/01/Statistics-in-Data-Science-Interview-Questions.pdf
- Youssef Hosni, public GitHub question collection: https://github.com/youssefHosni/Data-Science-Interview-Questions-Answers
- NIST/SEMATECH e-Handbook of Statistical Methods (reference for statistical definitions and methods): https://www.itl.nist.gov/div898/handbook/

Attribution note: This document does not claim that all 60 questions were asked at a named employer. Questions are original or adapted practice questions based on common foundational interview themes unless a specific public source is explicitly identified for a question. No employer-specific claim is made where source evidence is insufficient.

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STATISTICS INTERVIEW PREPARATION QUESTION BANK

LEVEL 2 — INTERMEDIATE

64 Questions | Solved Numericals | Python Practice | Selective Follow-ups

Data Analytics • Data Science • Statistical Inference

How to Use This Question Bank

- Attempt each question before reading the answer. Explain your reasoning as you would in an interview.
- For numerical problems, show the formula, substitute values, calculate the result, and interpret it in context.
- Python examples are separate from explanations. Run them in an environment with the stated libraries installed.
- Difficulty stars indicate the expected reasoning level:
  ★☆☆ foundational,
  ★★☆ intermediate,
  ★★★ more challenging.
- Follow-up questions are included selectively. Each retained follow-up has its answer immediately underneath it.

Level 2 Coverage Map

Topic	Coverage
Probability distributions	Random variables, binomial, Poisson, normal distribution, expected value
Sampling distributions and CLT	Sampling distribution, standard error, CLT, sample proportions
Estimation and confidence intervals	Point vs interval estimates, margin of error, z and t intervals
Hypothesis testing	Null/alternative hypotheses, p-values, α, Type I/II errors, power, tails, practical significance
Tests for means	One-sample t-test, paired vs independent samples, Welch's t-test, assumptions
Categorical tests and ANOVA	Chi-square goodness-of-fit/independence, expected counts, ANOVA and post-hoc comparisons
Applied reporting and Python	Business interpretation, confidence intervals, SciPy test examples, A/B testing outline

SECTION A — INTERMEDIATE STATISTICS

Q1. Explain the difference between a probability distribution and a frequency distribution. [★☆☆]

ANSWER

A frequency distribution summarizes observed data. A probability distribution describes the probabilities of possible outcomes of a random variable.

Q2. What is a random variable? Differentiate discrete and continuous random variables. [★☆☆]

ANSWER

A random variable assigns a number to each outcome of a random process. Discrete variables have countable outcomes; continuous variables can take any value in an interval.

Q3. A fair die is rolled. Define X as the number shown. What is P(X ≥ 5)? [★★☆]

ANSWER

The favourable outcomes are 5 and 6, out of 6 equally likely outcomes.

SOLVED NUMERICAL

P(X ≥ 5) = favourable outcomes ÷ total outcomes
= 2 ÷ 6 = 1/3 ≈ 0.3333 = 33.33%
Interpretation: The chance of rolling a 5 or 6 is about 33.33%.

Q4. What are expected value and variance of a discrete random variable? [★★☆]

ANSWER

Expected value is the probability-weighted average of outcomes. Variance is the probability-weighted average of squared distances from the expected value.

FOLLOW-UP QUESTION

What does the independence assumption mean here?

Answer: One lead's conversion outcome does not change the conversion probability of another lead.

Q5. A sales lead converts with probability 0.2. If each lead converts independently, what is the probability that exactly 2 of 5 leads convert? [★★★]

ANSWER

This follows a binomial model with n = 5, p = 0.2, and k = 2.

SOLVED NUMERICAL

Binomial formula: P(X = k) = C(n,k) p^k (1−p)^(n−k)
Substitute: P(X=2) = C(5,2)(0.2)^2(0.8)^3
Calculate: 10 × 0.04 × 0.512 = 0.2048 = 20.48%
Interpretation: The probability that exactly 2 of the 5 leads convert is 20.48%, under the independence and constant-probability assumptions.

Q6. When is a binomial distribution appropriate? [★ ☆ ☆]

ANSWER

Use it when there is a fixed number of independent trials, two outcomes per trial, and a constant probability of success.

Q7. A call centre receives an average of 4 escalations per hour. Under a Poisson model, what is the probability of exactly 2 escalations in an hour? [★ ★ ★]

ANSWER

Use a Poisson distribution with rate $\lambda = 4$.

SOLVED NUMERICAL

Formula: $P(X=k) = e^{(-\lambda)} \lambda^k \div k!$
Substitute: $P(X=2) = e^{(-4)} \times 4^2 \div 2!$
Calculate: $e^{(-4)} \times 16 \div 2 \approx 0.1465 = 14.65\%$
Interpretation: If the Poisson assumptions are reasonable, the chance of exactly two escalations in an hour is about 14.65%.

FOLLOW-UP QUESTION

What does λ represent in the Poisson model?

Answer: λ is the expected event count in the specified interval; here it is 4 escalations per hour.

Q8. How does a Poisson distribution differ from a binomial distribution? [★ ★ ☆]

ANSWER

Binomial counts successes in a fixed number of trials. Poisson models counts of events in a fixed interval when events occur at a stable average rate.

Q9. What does the normal distribution describe, and what are its key properties? [★ ☆ ☆]

ANSWER

It is a symmetric, bell-shaped continuous distribution. Mean, median, and mode are equal, and the total area under the curve is 1.

Q10. If X is normally distributed with mean 100 and SD 15, calculate the Z-score for X = 130. [★ ★ ☆]

ANSWER

The Z-score tells how many standard deviations the value lies from the mean.

SOLVED NUMERICAL

Formula: $Z = (x - \mu) \div \sigma$
Substitute: $Z = (130 - 100) \div 15$
Calculate: $Z = 30 \div 15 = 2$
Interpretation: 130 is two standard deviations above the mean.

Q11. What is the difference between a standard deviation and a standard error? [★ ☆ ☆]

ANSWER

Standard deviation describes spread among individual observations. Standard error describes the variability of a sample statistic across repeated samples.

Q12. A population has SD 18. For a random sample of 81, calculate the standard error of the sample mean. [★★☆]

ANSWER

For a sample mean, SE = population SD ÷ square root of sample size.

SOLVED NUMERICAL

Formula: SE = σ ÷ √n
Substitute: SE = 18 ÷ √81
Calculate: SE = 18 ÷ 9 = 2
Interpretation: The standard error of the sample mean is 2 units, assuming the sampling design supports this formula.

Q13. If sample size increases from 25 to 100 while population SD stays constant, how does standard error change? [★★☆]

ANSWER

Because SE is proportional to 1/√n, quadrupling sample size halves the standard error.

SOLVED NUMERICAL

Initial SE = σ ÷ √25 = σ ÷ 5
New SE = σ ÷ √100 = σ ÷ 10
Comparison: New SE ÷ initial SE = (σ/10) ÷ (σ/5) = 0.5
Interpretation: Standard error is halved, not reduced to one-quarter.

FOLLOW-UP QUESTION
Why does quadrupling sample size halve the standard error?

Answer: Standard error is proportional to the inverse square root of sample size: √4 = 2.

Q14. State the Central Limit Theorem and explain why analysts use it. [★★☆]

ANSWER

Under suitable conditions, the sampling distribution of the sample mean becomes approximately normal as sample size grows, even when the population is not normal. It supports inference about means.

Q15. Does the Central Limit Theorem say the raw data become normally distributed as sample size grows? [★☆☆]

ANSWER

No. It concerns the sampling distribution of a statistic, commonly the sample mean, not the distribution of the raw observations.

Q16. What conditions should you consider before relying on the CLT? [★ ★ ★]

ANSWER

Consider random or representative sampling, independence or dependence adjustments, finite variance, and whether sample size is adequate given skewness and outliers.

Q17. What is a sampling distribution? [★ ☆ ☆]

ANSWER

It is the probability distribution of a statistic, such as a mean or proportion, across repeated samples of the same size from the same population.

Q18. A sample proportion is 0.60 from n = 400. Calculate its approximate standard error. [★ ★ ★]

ANSWER

For a sample proportion, SE ≈ √[p̂(1−p̂)/n].

SOLVED NUMERICAL

Formula: $SE(\hat{p}) = \sqrt{[\hat{p}(1-\hat{p}) \div n]}$
Substitute: $\sqrt{[0.60 \times 0.40 \div 400]}$
Calculate: $\sqrt{(0.24 \div 400)} = \sqrt{0.0006} \approx 0.0245$
Interpretation: The estimated standard error is about 0.0245, or 2.45 percentage points.

Q19. What is a point estimate, and how does it differ from an interval estimate? [★ ☆ ☆]

ANSWER

A point estimate is one best estimate of a parameter. An interval estimate gives a range of plausible values with a stated confidence level.

Q20. Interpret a 95% confidence interval for average order value of ₹820 to ₹900. [★ ★ ☆]

ANSWER

Using the method repeatedly, about 95% of intervals constructed this way would contain the true population mean. It is not strictly a 95% probability statement about this already calculated fixed interval.

FOLLOW-UP QUESTION

Does a 95% confidence interval mean there is a 95% probability that this fixed interval contains the mean?

Answer: In the frequentist interpretation, the method captures the true mean in about 95% of intervals over repeated samples.

Q21. A sample mean is 50, population SD is 10, and n = 100. Construct an approximate 95% confidence interval using z* = 1.96. [★ ★ ★]

ANSWER

The margin of error is z* × SE; then add and subtract it from the sample mean.

SOLVED NUMERICAL

Step 1 — Standard error: SE = σ ÷ √n = 10 ÷ √100 = 1
Step 2 — Margin of error: ME = 1.96 × 1 = 1.96
Step 3 — Confidence interval: 50 ± 1.96
Lower limit = 48.04; upper limit = 51.96
Interpretation: The approximate 95% confidence interval is (48.04, 51.96), assuming the known-SD z interval is appropriate.

Q22. What happens to confidence interval width when confidence level increases, all else equal? [★ ☆ ☆]

ANSWER

The interval becomes wider because a higher confidence level requires a larger critical value.

Q23. What is margin of error, and which factors affect it? [★ ☆ ☆]

ANSWER

Margin of error is the critical value multiplied by the standard error. It increases with confidence level and variability, and generally decreases as sample size increases.

Q24. When would you use a t interval instead of a z interval for a population mean? [★ ★ ☆]

ANSWER

Use a t interval when population SD is unknown and estimated from the sample, particularly for smaller samples when assumptions are reasonable.

Q25. A poll has $\hat{p}$ = 0.52 and n = 1,000. Calculate an approximate 95% confidence interval using z* = 1.96. [★ ★ ★]

ANSWER

Use the estimated proportion standard error and add/subtract 1.96 standard errors.

SOLVED NUMERICAL

SE = √[0.52 × 0.48 ÷ 1,000] = √0.0002496 ≈ 0.0158
Margin of error = 1.96 × 0.0158 ≈ 0.0310
Interval = 0.52 ± 0.0310 = (0.489, 0.551)
Interpretation: The approximate 95% interval is 48.9% to 55.1%, subject to sampling and normal-approximation conditions.

Q26. What is the difference between a parameter and an estimate? [★ ☆ ☆]

ANSWER

A parameter is a fixed but often unknown population quantity. An estimate is calculated from sample data to approximate it.

Q27. What is the purpose of hypothesis testing? [★ ☆ ☆]

ANSWER

It uses sample evidence to assess whether data are inconsistent with a specified null hypothesis, under stated assumptions.

Q28. Define the null and alternative hypotheses using a business example. [★ ☆ ☆]

ANSWER

For a delivery-time claim: H_0: the population mean delivery time is 30 minutes; H_1: the mean differs from 30 minutes. The

alternative may instead be one-sided if justified in advance.

Q29. What is a p-value? [★★☆]

ANSWER

The probability, assuming the null hypothesis and model assumptions are true, of obtaining a test statistic at least as extreme as the observed one in the direction(s) defined by the test.

Q30. A test returns p = 0.03 at α = 0.05. What is the decision? [★★☆]

ANSWER

Since $p < \alpha$, reject the null hypothesis at the 5% significance level. This does not measure the size or business importance of the effect.

31. A test returns p = 0.08 at α = 0.05. Can you conclude the null hypothesis is true? [★★☆]

ANSWER

No. Fail to reject the null hypothesis; the evidence is insufficient at the 5% level, but that does not prove the null is true.

FOLLOW-UP QUESTION

Can a small p-value tell you whether the effect is important to the business?

Answer: No. Consider the effect size, confidence interval, costs, and business context.

Q32. Differentiate statistical significance and practical significance. [★☆☆]

ANSWER

Statistical significance relates to compatibility with the null under a test. Practical significance asks whether the effect is large or valuable enough to matter in context.

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Q33. Explain Type I and Type II errors. [★☆☆]

ANSWER

Type I error is rejecting a true null hypothesis. Type II error is failing to reject a false null hypothesis.

Q34. What is statistical power, and how can it be increased? [★★☆]

ANSWER

Power is the probability of rejecting the null when a specified alternative is true. It can increase with larger sample size, larger true effect, lower variability, or a higher α, with trade-offs.

Q35. Why should the significance level be selected before looking at the results? [★★☆]

ANSWER

Pre-specifying α limits outcome-driven decision changes and reduces the temptation to manipulate conclusions after seeing the data.

Q36. What is a one-tailed test, and when is it appropriate? [★★☆]

ANSWER

It tests an effect in one pre-specified direction. Use it only when effects in the opposite direction are not part of the decision question; do not choose it after seeing the data.

FOLLOW-UP QUESTION

Can increasing power increase the Type I error rate?

Answer: Not if α is held fixed and the design is improved appropriately; raising α can increase power but also raises Type I error risk.

Q37. What is a two-tailed test? [★☆☆]

ANSWER

It tests for departures in either direction from the null value.

Q38. For a two-sided z-test, z = 2.10 and α = 0.05. Using critical values ±1.96, what is the decision? [★★☆]

ANSWER

Since 2.10 is greater than 1.96, the statistic lies in the rejection region.

SOLVED NUMERICAL

Decision rule: Reject H_o if $z < -1.96$ or $z > 1.96$
Observed $z = 2.10$; because $2.10 > 1.96$, reject H_o at $\alpha = 0.05$
Interpretation: The result is statistically significant at the 5% level for the stated two-sided test.

Q39. What is the difference between statistical assumptions and the research hypothesis? [★★☆]

ANSWER

The research hypothesis is the claim being tested. Assumptions are conditions about the sampling process or data model needed for the test's results to be valid.

Q40. What should you do if a result is not statistically significant but the estimated effect looks large? [★★★]

ANSWER

Review uncertainty, sample size, confidence interval, data quality, and assumptions. Avoid claiming a confirmed effect, but do not equate non-significance with no effect.

Q41. When comparing a sample mean to a known benchmark with known population SD, which common test may be used? [★☆☆]

ANSWER

A one-sample z-test may be appropriate if the population SD is known and sampling/distribution assumptions are reasonable.

Q42. A sample has n = 25, mean = 104, and sample SD = 10. Test H_o: μ = 100 using a one-sample t statistic. [★★★]

ANSWER

Use t = (sample mean − hypothesized mean) ÷ (sample SD/√n).

SOLVED NUMERICAL

Formula: $t = (\bar{x} - \mu_o) \div (s \div \sqrt{n})$

Substitute: $t = (104 - 100) \div (10 \div \sqrt{25})$

Calculate: $t = 4 \div 2 = 2.00$; degrees of freedom = n − 1 = 24

Interpretation: The test statistic is t = 2.00. The final decision depends on the pre-specified tail(s), α, and the t distribution with 24 degrees of freedom.

Q43. What is the difference between paired and independent samples? [★☆☆]

ANSWER

Paired samples contain linked observations, such as before/after measurements on the same people. Independent samples come from unrelated groups.

FOLLOW-UP QUESTION

Why might a large sample produce a statistically significant but tiny effect?

Answer: Large samples reduce standard errors, making small effects easier to distinguish from zero.

Q44. A company measures employee productivity before and after training for the same employees. Which design is this? [★☆☆]

ANSWER

It is a paired design because each employee contributes a linked before-and-after measurement.

Q45. When is Welch's t-test preferred to the pooled independent-samples t-test? [★★☆]

ANSWER

Welch's t-test is useful when comparing independent means without assuming equal population variances, especially when group sizes and variances differ.

46. What assumptions should be checked for a t-test? [★★★]

ANSWER

Check independence and sampling design, influential outliers, and whether the distribution of observations or paired differences is suitable. For a pooled t-test, also consider equal variances.

Q47. Two independent groups have means 72 and 68. What additional information is needed to test whether their means differ? [★★☆]

ANSWER

You need sample sizes, group variability, study design, independence, the chosen significance level, and whether equal variances can reasonably be assumed.

Q48. What does a confidence interval add to a hypothesis test? [★☆☆]

ANSWER

It gives a range of plausible effect values and shows precision, helping interpret both magnitude and uncertainty rather than only a reject/fail-to-reject decision.

FOLLOW-UP QUESTION

Why does Welch's test not require equal population variances?

Answer: It uses a standard error and degrees-of-freedom approximation that accommodates unequal variances.

Q49. A 95% CI for a mean difference is (1.2, 5.8). What does this suggest for a two-sided test at 5%? [★★☆]

ANSWER

Because the interval excludes zero, it corresponds to rejecting a zero-difference null at the 5% level under the same method and assumptions.

Q50. What is a chi-square goodness-of-fit test used for? [★☆☆]

ANSWER

It compares observed counts in categories with expected counts from a specified distribution or proportions.

Q51. A category has observed count 60 and expected count 50. What is its contribution to the chi-square statistic? [★★☆]

ANSWER

For one cell, contribution = (observed − expected)² ÷ expected.

SOLVED NUMERICAL

Formula: Contribution = $(O - E)^2 \div E$
Substitute: $(60 - 50)^2 \div 50$
Calculate: $100 \div 50 = 2$
Interpretation: This category contributes 2 to the overall chi-square statistic; all categories' contributions are added.

Q52. When is a chi-square test of independence appropriate? [★★☆]

ANSWER

It tests whether two categorical variables are associated using counts in a contingency table, provided observations are independent and expected counts are adequate.

Q53. Why can very small expected counts be a problem in a chi-square test? [★★☆]

ANSWER

The chi-square approximation may be inaccurate. Consider combining categories only when substantively justified or using an exact/alternative method.

FOLLOW-UP QUESTION

What is the difference between goodness-of-fit and independence tests?

Answer: Goodness-of-fit compares one categorical variable with expected proportions; independence tests association between two categorical variables.

Q54. What is ANOVA used for? [★ ☆ ☆]

ANSWER

ANOVA tests whether there is evidence that at least one of three or more population means differs; a significant result does not identify which groups differ.

Q55. If an ANOVA result is significant, can you conclude every group mean differs? [★ ★ ☆]

ANSWER

No. It indicates that not all means are equal; post-hoc comparisons with suitable error control are needed to identify differences.

Q56. What are common assumptions of one-way ANOVA? [★ ★ ★]

ANSWER

Independent observations, approximately normal residuals within groups, and equal population variances for the classical test. Robust alternatives may be appropriate when assumptions fail.

FOLLOW-UP QUESTION

What should follow a significant omnibus ANOVA result?

Answer: Use suitable post-hoc comparisons with multiplicity control to identify which groups differ.

Q57. A product team compares conversion rates for two independent groups. What test family may be suitable? [★ ★ ☆]

ANSWER

A two-proportion test or a chi-square test of independence may be suitable for a binary conversion outcome, depending on the design and assumptions.

Q58. How would you explain a p-value of 0.001 to a non-technical manager? [★ ★ ☆]

ANSWER

If the null hypothesis and assumptions were true, results this extreme or more extreme would occur about 0.1% of the time. It is not the probability that the null is true.

FOLLOW-UP QUESTION

What is the key interpretation limit of a p-value?

Answer: It is calculated assuming the null and model assumptions, and is not the probability that the null hypothesis is true.

Q59. What is the danger of running many tests and reporting only significant results? [★ ★ ☆]

ANSWER

The chance of false positives increases. Pre-specify primary outcomes and consider suitable multiple-comparison procedures.

FOLLOW-UP QUESTION

Why is an A/B test usually stronger for causal claims than a simple before-and-after comparison?

Answer: Random assignment helps balance confounders; before-and-after comparisons can be affected by seasonality, trends, and other changes.

Q60. Python: how do you calculate a confidence interval for a mean using a t distribution? [★★☆]

ANSWER

Use the sample mean, sample standard error, and a t critical value with n−1 degrees of freedom, or use a validated statistical library.

PYTHON CODE

```
import numpy as np
from scipy import stats

x = np.array([12, 15, 18, 20, 25], dtype=float)
n = len(x)
mean = x.mean()
se = stats.sem(x)
ci = stats.t.interval(0.95, df=n-1, loc=mean, scale=se)
print(mean, ci)
```

Q61. Python: how can you run a one-sample t-test against a benchmark of 100? [★★☆]

ANSWER

Use scipy.stats.ttest_1samp and specify the hypothesized population mean.

PYTHON CODE

```
from scipy import stats

values = [104, 98, 102, 110, 96, 105]
result = stats.ttest_1samp(values, popmean=100)
print('t statistic:', result.statistic)
print('p-value:', result.pvalue)
```

Q62. What should be included in a statistical test report for a business stakeholder? [★★★]

ANSWER

State the business question, hypotheses, method, sample size, key assumptions, estimate and confidence interval, test statistic/p-value, limitations, and practical recommendation.

Q63. A manager says p > 0.05 means there is no effect. How would you respond? [★★☆]

ANSWER

A non-significant result means the data did not provide sufficient evidence against the null at that threshold; the confidence interval and study power help show what effects remain plausible.

Q64. Final scenario: an online retailer wants to know whether a redesigned checkout increased conversion. Outline the analysis. [★★★]

ANSWER

Define conversion and the target population, randomize users if possible, pre-specify the metric and hypothesis, calculate sample size/power, check data quality, compare proportions with an appropriate test and interval, examine practical impact and guardrail metrics, and report limitations.

Quick Interview Checklist

- State the business question and identify the parameter or metric.
- Check data quality, sampling design, independence, and test assumptions.
- Distinguish standard deviation from standard error.
- Interpret confidence intervals and p-values correctly; do not claim that p-value is the probability the null is true.
- Report effect size and practical importance alongside statistical significance.
- Choose tests based on outcome type, number of groups, pairing, sample design, and assumptions.
- Document limitations and translate results into a clear business recommendation.

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STATISTICS INTERVIEW PREPARATION QUESTION BANK

LEVEL 3 — ADVANCED

61 Advanced Questions | Solved Numericals | Python Practice | Selective Follow-ups

Data Analytics • Data Science • Regression & Model Diagnostics

How to Use This Question Bank

- Attempt each question before reading the answer and explain the reasoning clearly.
- For numericals, show the formula, substitution, calculation, and interpretation.
- Python examples are separated into monospaced blocks. They assume the relevant libraries and input variables are available.
- Difficulty stars indicate expected reasoning:
  ★☆☆ foundational,
  ★★☆ intermediate,
  ★★★ advanced.
- Follow-ups are selective. Each follow-up answer appears immediately beneath its question.

Level 3 Coverage Map

Topic	Coverage
Regression foundations	Simple and multiple regression, coefficients, residuals, R^2 and adjusted R^2
Model diagnostics	OLS assumptions, multicollinearity, VIF, heteroscedasticity, autocorrelation
Influence and interpretation	Outliers, leverage, Cook's distance, confidence intervals, interactions
Feature design and causal caution	Categorical encoding, omitted-variable bias, prediction versus causation
Model validation	Train/validation/test splits, cross-validation, overfitting, leakage, regularization
Predictive evaluation	MAE, RMSE, MAPE limitations, baseline comparisons
Advanced analytics scenarios	Simpson's paradox, residual diagnostics, business reporting
Python workflow	Scikit-learn regression, metrics, VIF with statsmodels

SECTION A — ADVANCED APPLIED STATISTICS

Q1. Differentiate simple linear regression and multiple linear regression. [★ ☆ ☆]

ANSWER

Simple linear regression uses one predictor; multiple linear regression uses two or more predictors to estimate the outcome.

Q2. In the model $\hat{y} = 12 + 3x$, interpret the intercept and slope. [★ ☆ ☆]

ANSWER

The intercept predicts y = 12 when x = 0. The slope means the predicted outcome increases by 3 units for each one-unit increase in x, within the model's valid range.

Q3. A regression model is $\hat{y} = 20 + 4x$. Predict y when x = 7. [★ ☆ ☆]

ANSWER

Substitute x = 7 into the fitted equation.

SOLVED NUMERICAL

Formula: $\hat{y} = \beta_0 + \beta_1 x$
Substitute: $\hat{y} = 20 + 4(7)$
Calculate: $\hat{y} = 20 + 28 = 48$
Interpretation: The model predicts an outcome of 48 when x = 7.

Q4. What is the difference between correlation and regression? [★ ☆ ☆]

ANSWER

Correlation measures the strength and direction of association between variables. Regression models an outcome as a function of predictors and can support prediction; neither alone proves causation.

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Q5. Interpret $R^2 = 0.72$ in a model predicting monthly sales. [★ ★ ☆]

ANSWER

The model explains 72% of the observed variation in sales around the sample mean, under the fitted model. It does not mean predictions are 72% accurate or that predictors cause sales.

Q6. What is adjusted R^2, and why can it be useful? [★ ★ ☆]

ANSWER

Adjusted R^2 accounts for the number of predictors relative to sample size and can discourage adding predictors that contribute little explanatory value.

Q7. A model has SSE = 120 and SST = 500. Calculate R^2. [★ ★ ☆]

ANSWER

$R^2 = 1 - SSE/SST$.

SOLVED NUMERICAL

Formula: $R^2 = 1 - SSE \div SST$
Substitute: $R^2 = 1 - 120 \div 500$
Calculate: $R^2 = 1 - 0.24 = 0.76$
Interpretation: The model accounts for 76% of the sample variation relative to the mean-only baseline.

FOLLOW-UP QUESTION
What does $R^2 = 0.76$ not tell you?
Answer: It does not establish causation, guarantee out-of-sample accuracy, or show that all model assumptions are satisfied.

Q8. What does a residual represent in regression? [★☆☆]

ANSWER
A residual is the observed outcome minus the model's predicted outcome for an observation.

Q9. Observed sales are 145 and predicted sales are 132. Calculate the residual. [★☆☆]

ANSWER
Residual = observed − predicted.

SOLVED NUMERICAL
Formula: $e = y - \hat{y}$
Substitute: $e = 145 - 132$
Calculate: $e = 13$
Interpretation: The model underpredicted this observation by 13 units.

Q10. What assumptions are commonly considered for ordinary least squares regression inference? [★★★]

ANSWER
Common considerations include a suitable functional form, independent errors, constant error variance, no perfect multicollinearity, and appropriate error-distribution assumptions for small-sample inference. Exogeneity is central for unbiased coefficient interpretation.

Q11. What is multicollinearity? [★☆☆]

ANSWER
It occurs when predictors are strongly linearly related, making individual coefficient estimates unstable and their standard errors larger.

Q12. What is the Variance Inflation Factor (VIF)? [★★☆]

ANSWER
VIF quantifies how much a coefficient's variance is inflated by linear relationships between that predictor and the other predictors. Larger values indicate more collinearity concerns, but no single cutoff applies universally.

Q13. For a predictor with $R^2_j = 0.80$ when regressed on other predictors, calculate its VIF. [★★☆]

ANSWER
$VIF = 1/(1 - R^2_j)$.

SOLVED NUMERICAL

Formula: VIF = $1 \div (1 - R^2_j)$
Substitute: VIF = $1 \div (1 - 0.80)$
Calculate: VIF = $1 \div 0.20 = 5$
Interpretation: This predictor's coefficient variance is inflated by a factor of 5 relative to the no-collinearity case, given the standard VIF definition.

FOLLOW-UP QUESTION

Does VIF = 5 automatically mean a predictor must be removed?

Answer: No. Treat it as a diagnostic signal and consider the goal, coefficient stability, domain knowledge, and alternative specifications.

Q14. How can you investigate multicollinearity? [★★☆]

ANSWER

Inspect predictor correlations, VIFs, coefficient instability across models, and domain overlap. Possible responses include combining variables, removing redundant predictors, collecting more data, or using regularization when prediction is the goal.

Q15. What is heteroscedasticity? [★☆☆]

ANSWER

It means the variance of the regression errors changes across predictor levels or fitted values, violating the constant-variance assumption.

Q16. What does a funnel shape in a residual-versus-fitted plot suggest? [★★☆]

ANSWER

It may indicate heteroscedasticity: residual spread grows or shrinks with fitted values.

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17. How can heteroscedasticity affect inference and what can you do? [★★★]

ANSWER

Ordinary least squares coefficients may remain unbiased under suitable exogeneity conditions, but conventional standard errors and tests can be unreliable. Consider heteroscedasticity-robust standard errors, transformations, or a better-specified model.

FOLLOW-UP QUESTION

Does heteroscedasticity automatically bias OLS coefficients?

Answer: Not by itself if exogeneity holds, but conventional standard errors and inference can become unreliable.

Q18. What is autocorrelation in regression errors? [★☆☆]

ANSWER

It means errors are correlated across ordered observations, often over time. It can make conventional standard errors and tests unreliable.

Q19. Why can random train/test splitting be inappropriate for time-series data? [★★☆]

ANSWER

It may leak future information into training and violate the time order. Use chronological splits or rolling-origin validation.

Q20. What is an influential observation? [★☆☆]

ANSWER

An observation that substantially changes fitted coefficients or predictions when included or removed. Influence depends on residual size and leverage, not only extremeness.

Q21. Differentiate outlier, leverage, and influence. [★★☆]

ANSWER

An outlier has an unusual outcome relative to the model; a high-leverage point has unusual predictor values; an influential point materially changes the fitted model.

Q22. What is Cook's distance used for? [★★☆]

ANSWER

It summarizes how much the fitted regression changes when an observation is omitted. It is a diagnostic flag, not an automatic deletion rule.

FOLLOW-UP QUESTION

Can a large residual alone identify an influential point?

Answer: No. Influence also depends on leverage and how much the fitted model changes when the observation is omitted.

Q23. A point has high leverage but a small residual. Must it be removed? [★★☆]

ANSWER

No. High leverage alone does not prove the point is erroneous or influential. Validate it and assess its effect on the fitted model.

Q24. What is the difference between statistical and practical significance in regression? [★☆☆]

ANSWER

Statistical significance evaluates evidence against a zero or specified coefficient under assumptions. Practical significance evaluates the size, cost, and business value of the effect.

Q25. A coefficient is 2.4 with standard error 0.8. Calculate the approximate t-statistic for testing coefficient = 0. [★★☆]

ANSWER

t = estimate ÷ standard error.

SOLVED NUMERICAL

Formula: t = (estimate − hypothesized value) ÷ SE
Substitute: t = (2.4 − 0) ÷ 0.8

Calculate: t = 3.0
Interpretation: The coefficient is three standard errors above zero. The p-value depends on degrees of freedom and whether the test is one- or two-sided.

Q26. How do you interpret a 95% confidence interval for a regression coefficient of (0.5, 4.3)? [★★☆]

ANSWER

Under the model and interval method, the data are compatible with coefficient values from 0.5 to 4.3 at the stated confidence procedure. Since zero is excluded, the corresponding two-sided test of zero would be significant at 5% under the same assumptions.

Q27. What is an interaction term in multiple regression? [★★☆]

ANSWER

It allows the effect of one predictor to depend on the value of another predictor.

Q28. A model includes Spend = $\beta_0 + \beta_1$AdSpend + β_2Region + β_3(AdSpend×Region). What does β_3 represent? [★★★]

ANSWER

For the coding used, β_3 represents how the AdSpend slope differs for the coded Region group relative to the reference group.

Q29. Why should categorical variables often be encoded with indicator variables? [★☆☆]

ANSWER

Regression needs numerical design-matrix columns. Indicator variables represent category membership while a reference category avoids perfect redundancy in standard coding.

FOLLOW-UP QUESTION

Why is the reference category important for interpreting an interaction model?

Answer: The main-effect coefficients are interpreted relative to the reference coding; changing the reference changes coefficient interpretation but not fitted values.

Q30. What is a dummy-variable trap? [★★☆]

ANSWER

Including an intercept and a full set of indicators for every category creates perfect multicollinearity. Usually one category is omitted as the reference.

Q31. How can omitted-variable bias arise? [★★★]

ANSWER

If an omitted variable affects the outcome and is correlated with an included predictor, the included coefficient may absorb part of the omitted variable's effect.

Q32. What is the difference between prediction and causal inference? [★☆☆]

ANSWER

Prediction targets accurate forecasts for new cases. Causal inference targets the effect of an intervention and requires a design or assumptions that address confounding.

33. Why can a high R^2 still accompany poor predictions on new data? [★★☆]

ANSWER

The model may overfit, exploit leakage, or fail under distribution shift. Evaluate held-out or time-appropriate validation performance and compare against a baseline.

FOLLOW-UP QUESTION

Can a high R^2 prove a model is good for deployment?

Answer: No. Check leakage, held-out performance, stability, calibration where relevant, and whether the model meets the business objective.

Q34. What is overfitting in regression? [★☆☆]

ANSWER

A model fits noise or peculiarities of the training data and generalizes poorly to unseen data.

Q35. Differentiate training, validation, and test sets. [★☆☆]

ANSWER

Training data fit model parameters; validation data support tuning and model selection; the test set provides a final, preferably one-time, assessment of generalization.

Q36. What is regularization, and how do Ridge and Lasso differ? [★★☆]

ANSWER

Regularization penalizes model complexity. Ridge uses an L2 penalty and shrinks coefficients; Lasso uses an L1 penalty and can set some coefficients exactly to zero.

Q37. What is a key limitation of using Lasso for causal interpretation? [★★★]

ANSWER

Lasso is primarily a predictive/selection method; selection can distort conventional inference, and dropping confounders can bias causal estimates. Use a method designed for the causal objective.

Q38. Calculate MAE for actual values [10, 12, 15] and predictions [12, 11, 13]. [★★☆]

ANSWER

MAE is the mean of the absolute prediction errors.

SOLVED NUMERICAL

Absolute errors: $|10-12| = 2$; $|12-11| = 1$; $|15-13| = 2$
Formula: MAE = sum of absolute errors ÷ number of observations
Calculate: MAE = $(2 + 1 + 2) \div 3 = 5 \div 3 \approx 1.67$
Interpretation: Predictions differ from actual values by about 1.67 units on average in absolute terms.

Q39. Calculate RMSE for actual [10, 12, 15] and predicted [12, 11, 13]. [★★★]

ANSWER

RMSE is the square root of the mean squared prediction error.

SOLVED NUMERICAL

Errors: 2, −1, 2
Squared errors: 4, 1, 4
Formula: RMSE = √(sum of squared errors ÷ n)
Calculate: RMSE = √[(4 + 1 + 4) ÷ 3] = √3 ≈ 1.73
Interpretation: RMSE penalizes larger errors more strongly than MAE because errors are squared.

FOLLOW-UP QUESTION

Why can RMSE exceed MAE?

Answer: RMSE squares errors before averaging and taking a square root, so larger errors receive more weight; RMSE is at least MAE for the same observations.

Q40. Why should MAE and RMSE be compared with the outcome's scale? [★☆☆]

ANSWER

Their numerical size has meaning in the outcome's units; whether an error is acceptable depends on the business context and baseline.

Q41. What is MAPE, and when can it be problematic? [★★☆]

ANSWER

MAPE averages absolute percentage errors. It becomes undefined or unstable when actual values are zero or near zero and can behave asymmetrically.

Q42. A model's MAE is 5 units while a baseline model's MAE is 8. Calculate the relative MAE reduction. [★★☆]

ANSWER

Relative reduction compares the improvement with the baseline error.

SOLVED NUMERICAL

Formula: Relative reduction = (baseline MAE − model MAE) ÷ baseline MAE × 100%
Substitute: (8 − 5) ÷ 8 × 100%
Calculate: 3 ÷ 8 × 100% = 37.5%
Interpretation: The model reduces MAE by 37.5% relative to the baseline on the evaluated data.

Q43. What is the bias-variance trade-off? [★★☆]

ANSWER

Very simple models may underfit with high bias; very flexible models may overfit with high variance. Model selection balances these risks using validation.

Q44. What is data leakage in model development? [★ ☆ ☆]

ANSWER

Leakage occurs when training or model selection uses information unavailable at prediction time, producing overly optimistic evaluation.

Q45. Why should preprocessing be fitted inside each cross-validation training fold? [★ ★ ★]

ANSWER

Fitting preprocessing on all data lets information from validation folds influence training. Fit transformations within each fold to avoid leakage.

Q46. What is k-fold cross-validation? [★ ☆ ☆]

ANSWER

Split data into k folds; train on k−1 folds and validate on the remaining fold, repeating until each fold has served as validation. Aggregate the results.

FOLLOW-UP QUESTION

Why should time-series validation preserve chronology?

Answer: A deployment model predicts the future from the past; random splitting can let future patterns leak into training.

Q47. When might stratified cross-validation be useful? [★ ★ ☆]

ANSWER

For classification, it helps preserve approximate class proportions across folds, particularly when classes are imbalanced.

Q48. What is Simpson's paradox, and how can it affect business analysis? [★ ★ ★]

ANSWER

A trend seen in separate groups can reverse when groups are combined because group sizes or confounders differ. Inspect relevant segments and causal structure before aggregating.

49. Why is correlation among predictors not the same as confounding? [★ ★ ★]

ANSWER

Multicollinearity concerns relationships among predictors and coefficient stability; confounding concerns a variable distorting an exposure-outcome relationship in causal analysis.

Q50. A sales model uses advertising spend and sales. Name two reasons a positive coefficient may not prove advertising caused sales. [★ ★ ☆]

ANSWER

Budget may be raised when demand is expected to rise, and other factors such as seasonality, promotions, or price may influence both spend and sales.

Q51. How would you explain heteroscedasticity to a business stakeholder? [★☆☆]

ANSWER

Prediction errors are more variable for some types or sizes of cases than others, so the model's uncertainty is not uniform.

Q52. What would you inspect in residual diagnostics before trusting a regression? [★★☆]

ANSWER

Residuals versus fitted values, residuals versus predictors/time, Q–Q plot where relevant, influential observations, variance patterns, and evidence of misspecification.

FOLLOW-UP QUESTION

How does Simpson's paradox suggest a practical reporting check?

Answer: Report the overall result and relevant segment-level results, and investigate differences in group composition before drawing conclusions.

Q53. What does a residual Q–Q plot help assess? [★★☆]

ANSWER

It compares residual quantiles with theoretical normal quantiles to assess whether residual normality is a reasonable approximation, mainly relevant to classical small-sample inference.

Q54. How can you handle missing predictor values without leakage? [★★☆]

ANSWER

Fit imputation using training data only within the modeling pipeline, then apply the learned imputer to validation/test data.

Q55. Python: fit a multiple linear regression model with scikit-learn and report test-set R^2. [★★☆]

ANSWER

Split the data first, fit only on training data, and evaluate on held-out test data.

PYTHON CODE

```
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
model = LinearRegression()
model.fit(X_train, y_train)
pred = model.predict(X_test)
print('Test R²:', r2_score(y_test, pred))
```

Q56. Python: calculate MAE and RMSE for predictions. [★☆☆]

ANSWER

Use scikit-learn's metrics and compare them with a baseline on the same test set.

PYTHON CODE

```
from sklearn.metrics import mean_absolute_error, mean_squared_error
import numpy as np

mae = mean_absolute_error(y_test, pred)
rmse = np.sqrt(mean_squared_error(y_test, pred))
print('MAE:', mae)
print('RMSE:', rmse)
```

Q57. Python: calculate VIF for a design matrix. [★★★]

ANSWER

Use statsmodels' variance_inflation_factor on the predictor matrix; ensure the design matrix and intercept handling are appropriate.

PYTHON CODE

```
import pandas as pd
import statsmodels.api as sm
from statsmodels.stats.outliers_influence import variance_inflation_factor

X_const = sm.add_constant(X)
vif = pd.DataFrame({
    'feature': X_const.columns,
    'VIF': [variance_inflation_factor(X_const.values, i)
            for i in range(X_const.shape[1])]
})
print(vif)
```

Q58. What is the difference between OLS coefficients and standardized coefficients? [★★☆]

ANSWER

OLS coefficients express the predicted outcome change per one-unit predictor change, holding other predictors constant. Standardized coefficients express changes in standard-deviation units and can aid scale comparison, with caveats.

Q59. Why can a coefficient change sign after adding a predictor? [★★★]

ANSWER

The original model may have omitted-variable bias or correlated predictors. Interpret the conditional model and inspect collinearity and model specification rather than assuming the change is an error.

FOLLOW-UP QUESTION

Why is VIF for the intercept often not useful for feature-collinearity interpretation?

Answer: The intercept is not a substantive predictor; focus interpretation on the actual explanatory variables.

Q60. What should a robust regression report include? [★★★]

ANSWER

State the objective, sample and split design, features, coefficients with uncertainty where appropriate, diagnostic findings, held-out metrics versus a baseline, limitations, and the business action supported.

Q61. Final case: a retailer wants to predict weekly sales from price, ad spend, promotions, and seasonality. Outline an analysis plan. [★★★]

ANSWER

Define the forecast horizon and target, build a chronological split, inspect missingness/outliers, encode categories and time features, fit a baseline and candidate models, diagnose residuals and leakage, compare MAE/RMSE on held-out weeks, examine stability and uncertainty, and communicate limitations. Do not claim causal effects from predictive coefficients alone.

Advanced Interview Checklist

- Separate predictive performance from causal interpretation.
- Inspect residual patterns, influential observations, multicollinearity, and potential misspecification.
- Use chronological validation for time-dependent data and prevent preprocessing leakage.
- Compare candidate models against a sensible baseline using metrics aligned with business costs.
- Interpret coefficients in their units and coding context; include uncertainty where appropriate.
- Treat diagnostics as evidence to investigate, not automatic rules to delete rows or features.
- Translate findings into a decision, caveats, and next steps.

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STATISTICS INTERVIEW PREPARATION QUESTION BANK

LEVEL 4 — INDUSTRY CASES & ADVANCED CHALLENGES

57 Case-Based Questions | Solved Numericals | Decision Frameworks | Selective Follow-ups

Data Analytics • Data Science • Experimentation • Business Decision-Making

How to Use This Question Bank

- Treat each prompt as a real interview case: clarify the decision, define the metric, identify the data and design, analyse uncertainty, and recommend an action.
- For numericals, show the formula, substitute values, calculate, and interpret the result in business terms.
- Do not confuse association with causation, statistical significance with business value, or an aggregate result with segment-level performance.
- Difficulty stars indicate expected reasoning: ★☆☆ foundational, ★★☆ intermediate, ★★★ advanced.
- Follow-up questions are selective. Each retained follow-up answer appears immediately below its question.

Level 4 Coverage Map

Topic	Coverage
Experimentation and A/B testing	Randomization, conversion lift, guardrails, MDE, sequential monitoring
Causal inference	Difference-in-differences, parallel trends, interrupted time series, confounding
Business KPI interpretation	Churn, cohort analysis, denominator changes, Simpson's paradox
Statistical decisions	Test selection, effect size, multiple testing, family-wise error, FDR
Time-series evaluation	Seasonality, autocorrelation, rolling-origin validation, baseline comparisons
Classification evaluation	Precision, recall, class imbalance, thresholds, calibration, ROC-AUC vs PR
Data quality and representation	Duplicate records, sampling composition, post-stratification weights
Executive communication	Guardrails, model limitations, scaling decisions, business value

SECTION A — INDUSTRY CASES & ADVANCED CHALLENGES

Q1. An online retailer sees conversion rise after a checkout redesign. What evidence would you need before claiming the redesign caused the increase? [★★☆]

ANSWER

Prefer a randomized A/B test with a pre-defined conversion metric, adequate sample size, consistent tracking, and a suitable comparison. A before-and-after increase alone can be caused by seasonality, traffic mix, or other changes.

Q2. Design a basic A/B test for a new product page. [★★☆]

ANSWER

Define the hypothesis and primary metric, randomize eligible users, determine sample size and duration, set guardrail metrics, verify instrumentation, analyse the difference with an appropriate interval/test, and report effect size and limitations.

Q3. Variant A converts 120 of 2,000 visitors; Variant B converts 150 of 2,000. Calculate each conversion rate and the absolute lift. [★★☆]

ANSWER

Conversion rate = conversions ÷ visitors. Absolute lift is B's rate minus A's rate.

SOLVED NUMERICAL

A rate = 120 ÷ 2,000 = 0.06 = 6.0%
B rate = 150 ÷ 2,000 = 0.075 = 7.5%
Absolute lift = 7.5% − 6.0% = 1.5 percentage points
Relative lift = (7.5% − 6.0%) ÷ 6.0% = 25%
Interpretation: B is higher by 1.5 percentage points, a 25% relative increase. Statistical uncertainty must still be assessed before concluding the change is reliable.

FOLLOW-UP QUESTION

Why should you report both percentage-point lift and relative lift?

Answer: Percentage points show the absolute rate change; relative lift shows the change relative to the baseline and can look large when the baseline is small.

Q4. Differentiate absolute lift and relative lift in an experiment. [★☆☆]

ANSWER

Absolute lift is the difference between rates in percentage points. Relative lift divides that difference by the baseline rate and expresses it as a percentage.

Q5. Why should a team choose one primary metric before an A/B test? [★★☆]

ANSWER

Pre-specifying the primary metric reduces selective reporting and makes the decision rule clearer. Guardrail metrics can still be monitored for unintended harm.

Q6. What is a guardrail metric in experimentation? [★☆☆]

ANSWER

A metric monitored to ensure a change does not harm an important area, such as refunds, latency, complaints, or retention, while improving the primary metric.

Q7. Why can repeatedly checking an ordinary p-value and stopping as soon as it is significant inflate false positives? [★★★]

ANSWER

Repeated unadjusted significance checks change the overall Type I error rate. Use a pre-planned sample size/duration or a valid sequential-testing method.

Q8. What is the difference between randomization and random sampling? [★★☆]

ANSWER

Randomization assigns units to treatment groups to support causal comparisons. Random sampling selects units from a population and can support representativeness; they serve different purposes.

FOLLOW-UP QUESTION

What is one valid alternative to fixed-horizon testing?

Answer: A pre-specified sequential-testing method can support interim monitoring while controlling error rates.

Q9. An A/B test randomizes users, but users share offers with each other. What issue can arise? [★★★]

ANSWER

Interference between users can violate the assumption that one user's treatment does not affect another user's outcome. Consider cluster or geo-level randomization or adjust the design.

Q10. What is a minimum detectable effect (MDE)? [★★☆]

ANSWER

It is the smallest effect a study is designed to detect with specified power and significance level. It should be tied to business value, not chosen only for convenience.

Q11. Why is a statistically significant uplift not automatically a business win? [★★☆]

ANSWER

The uplift may be too small to cover implementation or operating costs, or it may worsen guardrail metrics. Assess net value, uncertainty, and long-term effects.

Q12. A campaign costs ₹80,000 and generates ₹1,20,000 incremental contribution. Calculate net contribution after campaign cost and ROI on cost. [★★☆]

ANSWER

Net contribution = incremental contribution − campaign cost. ROI on cost = net contribution ÷ campaign cost.

SOLVED NUMERICAL

Net contribution = ₹1,20,000 − ₹80,000 = ₹40,000
ROI on cost = ₹40,000 ÷ ₹80,000 × 100% = 50%
Interpretation: The campaign produces ₹40,000 after campaign cost, equivalent to a 50% return on that cost, assuming the incremental contribution estimate is credible.

Q13. How would you evaluate a campaign when randomization is not possible? [★★★]

ANSWER

Consider a quasi-experimental approach such as difference-in-differences, matching, regression adjustment, or interrupted time series, while explicitly stating assumptions and possible confounding.

Q14. Explain difference-in-differences (DiD) in plain language. [★★☆]

ANSWER

Compare the before-to-after change in a treated group with the before-to-after change in a comparable untreated group. The causal interpretation relies on assumptions such as parallel trends absent treatment.

Q15. A treated region rises from 100 to 130 sales; a control region rises from 80 to 95. Calculate the DiD estimate. [★★★]

ANSWER

DiD = change in treated group − change in control group.

SOLVED NUMERICAL

Treated change = 130 − 100 = +30
Control change = 95 − 80 = +15
DiD = 30 − 15 = +15 sales
Interpretation: The estimated incremental change is 15 sales relative to the control trend, if the parallel-trends and other identification assumptions are credible.

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Q16. What is the parallel-trends assumption in DiD? [★★★]

ANSWER

In the absence of treatment, the treated and control groups would have followed similar average outcome trends. Pre-treatment trend checks can inform, but cannot prove, this assumption.

FOLLOW-UP QUESTION

Can pre-treatment trend similarity prove the DiD assumption?

Answer: No. It provides useful evidence but cannot prove the unobserved counterfactual trends would remain parallel.

Q17. When might interrupted time-series analysis be useful? [★★☆]

ANSWER

When an intervention occurs at a known time and there are enough observations before and after to estimate changes in level or trend, while considering seasonality, autocorrelation, and concurrent events.

Q18. A business reports strong December sales. How would you distinguish seasonality from a real improvement? [★★☆]

ANSWER

Compare with prior Decembers, model trend and seasonal patterns, account for trading days and promotions, and use a suitable baseline or control if available.

Q19. What is a confounder in an observational analysis? [★★☆]

ANSWER

A variable associated with both the exposure and outcome that can distort the estimated relationship if not appropriately handled. Association alone does not establish that a variable is a confounder in every design.

Q20. Why can controlling for every available variable make an analysis worse? [★★★]

ANSWER

Some variables may be mediators or colliders, or measured after treatment. Blind adjustment can introduce bias; choose covariates based on the causal question and design.

Q21. A subscription company sees churn fall from 8% to 6%. Calculate the absolute and relative reduction. [★★☆]

ANSWER

Absolute reduction = old rate − new rate. Relative reduction = absolute reduction ÷ old rate.

SOLVED NUMERICAL

Absolute reduction = 8% − 6% = 2 percentage points
Relative reduction = (8% − 6%) ÷ 8% × 100% = 25%
Interpretation: Churn fell by 2 percentage points, which is a 25% relative reduction compared with the original 8% rate.

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FOLLOW-UP QUESTION

Why is blind adjustment for all variables risky?

Answer: Adjusting for mediators, colliders, or post-treatment variables can introduce bias instead of removing it.

Q22. How should you report a business KPI when its denominator changes over time? [★★☆]

ANSWER

Report the numerator and denominator, define the metric consistently, compare rates rather than raw counts when appropriate, and investigate changes in population mix or eligibility.

Q23. What is cohort analysis, and why is it useful for retention? [★☆☆]

ANSWER

Group users by a shared starting event or period and track outcomes over time. It helps distinguish lifecycle patterns from changes in the mix of users acquired.

Q24. A dashboard shows revenue up 20%, but orders are flat. Name plausible explanations to investigate. [★★☆]

ANSWER

Average order value or product mix may have risen, prices may have changed, refunds or revenue recognition may differ, or tracking/period definitions may have changed.

Q25. What is Simpson's paradox, and how would you investigate it in a KPI dashboard? [★★★]

ANSWER

An aggregate trend can reverse within subgroups due to differences in group composition. Compare overall and segmented metrics, inspect denominators, and investigate confounding before recommending action.

Q26. How would you decide whether to use a t-test, chi-square test, or ANOVA? [★★☆]

ANSWER

Use a t-test for means in one or two groups under suitable assumptions, chi-square for categorical count relationships, and ANOVA for comparing means across three or more groups. Pairing, design, distribution, and alternatives also matter.

Q27. A team compares mean satisfaction across four independent branches. What test may be appropriate and what happens after a significant result? [★★☆]

ANSWER

One-way ANOVA may be appropriate if its assumptions are reasonable. If significant, use suitable post-hoc comparisons with multiplicity control to identify differences.

Q28. A test is significant with p = 0.001, but the effect is tiny. What should be reported? [★★☆]

ANSWER

Report the effect estimate, confidence interval, sample size, p-value, practical threshold, costs, and limitations. Statistical significance alone does not establish business value.

Q29. What is multiple-testing correction and when may it be needed? [★★★]

ANSWER

It controls the increased risk of false positives when many hypotheses are tested. The approach depends on whether the goal is family-wise error control, false discovery rate, or another pre-specified objective.

Q30. What is the difference between family-wise error rate and false discovery rate? [★★★]

ANSWER

Family-wise error rate is the probability of at least one false positive in a family of tests. False discovery rate is the expected proportion of false positives among rejected hypotheses, under its formal definition.

Q31. A data scientist tests 25 independent null hypotheses at $\alpha = 0.05$ without correction. What is the approximate chance of at least one false positive if all nulls are true? [★★★]

ANSWER

Assuming independent tests, probability of no false positives is $(1 - \alpha)^m$; subtract this from 1.

SOLVED NUMERICAL

P(at least one false positive) = 1 − (1 − 0.05)^25
= 1 − 0.95^25
≈ 1 − 0.2774 = 0.7226
Interpretation: The approximate family-wise probability is 72.3% under the stated independence and all-null assumptions.

FOLLOW-UP QUESTION

What does the 72.3% result assume?

Answer: It assumes 25 independent tests, each with a true null and Type I error probability of 5%.

Q32. What is effect size, and why report it with a p-value? [★☆☆]

ANSWER

Effect size quantifies magnitude, such as a mean difference, proportion difference, or standardized effect. It helps assess practical importance independently of sample size alone.

33. Explain the difference between a confidence interval and a prediction interval in regression. [★★★]

ANSWER

A confidence interval for the mean response describes uncertainty in the estimated average outcome at predictor values. A prediction interval covers an individual future outcome and is wider because it includes individual-level noise.

Q34. Why can a time-series regression have high R^2 but misleading significance tests? [★★★]

ANSWER

Serially correlated errors, omitted seasonality, common trends, or non-stationarity can invalidate conventional inference and create spurious relationships.

Q35. What is autocorrelation in a time series, and how can it affect model evaluation? [★★☆]

ANSWER

Autocorrelation means observations or errors are related across time. It violates independence assumptions for many standard procedures and requires time-aware diagnostics and validation.

Q36. What is a rolling-origin validation strategy? [★★☆]

ANSWER

Train on an initial historical window, evaluate on a later period, then move the cutoff forward and repeat. It mimics repeated real-world forecasting more faithfully than random splits.

FOLLOW-UP QUESTION

Which interval is usually wider: confidence interval for the mean response or prediction interval for an individual?

Answer: The prediction interval is wider because it includes individual outcome noise as well as uncertainty in the mean estimate.

Q37. A forecast model has MAE 12; a seasonal-naive baseline has MAE 10. What is your conclusion? [★★☆]

ANSWER

The candidate model is worse than the baseline on the evaluated data. Investigate the setup and consider the baseline unless another business objective justifies the candidate.

SOLVED NUMERICAL

Relative MAE change = (12 − 10) ÷ 10 × 100% = 20%

Interpretation: The candidate's MAE is 20% higher than the baseline, so it performs worse on this evaluation period.

Q38. How would you choose between MAE and RMSE for a business forecast? [★★☆]

ANSWER

Use MAE when average absolute error is easy to explain and robust to extreme errors; use RMSE when large errors should be penalized more. Match the metric to the cost of errors.

Q39. What is a prediction interval, and why is it useful for operational planning? [★★☆]

ANSWER

It gives a range for a future individual outcome under model assumptions. It helps plan capacity and risk rather than relying on a single point forecast.

Q40. What is the difference between uncertainty in a parameter estimate and variability in future observations? [★★★]

ANSWER

Parameter uncertainty concerns how precisely the model's underlying quantity is estimated. Future-observation variability also includes irreducible individual-level noise.

FOLLOW-UP QUESTION

Why can random cross-validation be unsuitable for forecasting?

Answer: It can train on future observations and validate on the past, creating leakage that would not be possible at deployment.

Q41. How would you assess whether a model is fair across customer segments? [★★★]

ANSWER

Evaluate performance and error rates by relevant segments, check representation and data quality, consider harms and legal context, and assess whether differences reflect legitimate needs or systematic disadvantage.

Q42. A model's overall accuracy is 95%, but it misses most positive cases in a rare-event setting. What should you inspect? [★★☆]

ANSWER

Inspect the confusion matrix, recall/sensitivity, precision, specificity, PR curve, threshold trade-offs, and class prevalence. Accuracy alone can hide poor minority-class detection.

Q43. How does changing a classification threshold affect precision and recall? [★★☆]

ANSWER

Lowering the threshold typically increases recall and may reduce precision; raising it typically increases precision and may reduce recall. The exact trade-off depends on score distributions.

Q44. Why can accuracy be misleading with class imbalance? [★☆☆]

ANSWER

A model can predict the majority class for almost every case and still achieve high accuracy while failing to detect the minority class of interest.

Q45. How would you choose a threshold for a fraud model? [★★★]

ANSWER

Estimate the costs of false positives and false negatives, review precision-recall trade-offs, consider operational capacity and policy constraints, and validate the threshold on data representative of deployment.

Q46. A fraud model flags 200 transactions; 150 are truly fraudulent. Calculate precision. [★★☆]

ANSWER

Precision = true positives ÷ all predicted positives.

SOLVED NUMERICAL

Formula: Precision = TP ÷ (TP + FP)
Substitute: Precision = 150 ÷ 200
Calculate: Precision = 0.75 = 75%
Interpretation: 75% of flagged transactions are truly fraudulent in this evaluation sample.

Q47. The model catches 150 of 300 truly fraudulent transactions. Calculate recall. [★★☆]

ANSWER

Recall = true positives ÷ all actual positives.

SOLVED NUMERICAL

Formula: Recall = TP ÷ (TP + FN)
Substitute: Recall = 150 ÷ 300
Calculate: Recall = 0.50 = 50%
Interpretation: The model detects half of all fraudulent transactions in the evaluation sample.

FOLLOW-UP QUESTION

Why can high overall accuracy conceal model failure?

Answer: When positive cases are rare, predicting the majority class can yield high accuracy while recall for the positive class remains poor.

Q48. What is calibration in a probabilistic model? [★ ★ ★]

ANSWER

Calibration means predicted probabilities align with observed event frequencies across cases or groups. A model can rank cases well yet be poorly calibrated.

49. What is the difference between ROC-AUC and precision-recall analysis? [★ ★ ★]

ANSWER

ROC-AUC summarizes ranking across thresholds using true-positive and false-positive rates. Precision-recall analysis focuses on positive predictive performance and is often more informative for rare positive classes.

Q50. A classifier has ROC-AUC = 0.90. Does that mean 90% of predictions are correct? [★ ★ ☆]

ANSWER

No. AUC measures ranking discrimination across thresholds; it is not classification accuracy or calibration.

FOLLOW-UP QUESTION

Can ROC-AUC alone determine the operational threshold?

Answer: No. Threshold selection depends on costs, capacity, precision-recall trade-offs, and the deployment context.

Q51. How would you communicate a model's limitations to an executive? [★ ★ ☆]

ANSWER

Explain the intended use, validation population and period, performance versus baseline, uncertainty, known failure modes, data drift risks, operational controls, and decisions the model should not make alone.

Q52. A dashboard KPI suddenly doubles overnight. What is your first investigation sequence? [★ ★ ☆]

ANSWER

Verify definitions and date filters, inspect source freshness and pipeline logs, check duplicates and joins, compare raw counts and denominators, segment the change, and confirm whether a real business event occurred.

Q53. How can duplicate records distort a conversion-rate analysis? [★ ☆ ☆]

ANSWER

Duplicates can inflate visitors, conversions, or both, biasing the rate. Define the unit of analysis and deduplicate using validated identifiers and event rules.

Q54. A sample contains 70% mobile users while the target population is 40% mobile. What concern arises? [★ ★ ☆]

ANSWER

The sample composition differs from the target population, so aggregate estimates may be biased if outcomes differ by device. Consider representative sampling, stratification, or justified weighting.

Q55. When can post-stratification weights help, and what are their limitations? [★★★]

ANSWER

Weights can align sample group shares to known population shares. They cannot fix unmeasured selection bias automatically and may increase variance, especially with extreme weights.

Q56. A dashboard shows a 10% overall improvement but a decline in the highest-value customer segment. What should you recommend? [★★★]

ANSWER

Validate metric definitions and sample sizes, inspect segment mix and uncertainty, quantify the high-value segment's business impact, and avoid relying on the aggregate alone before making a decision.

FOLLOW-UP QUESTION

What is the first question to ask when a dashboard KPI doubles overnight?

Answer: Ask whether the metric definition, data pipeline, filters, or source changed before assuming business performance truly doubled.

Q57. Final case: an executive asks whether a retention campaign worked, whether to scale it, and how confident you are. Give a structured analysis plan. [★★★]

ANSWER

Clarify the target population and retention outcome, define the evaluation window, prefer randomized holdout if feasible, estimate absolute and relative effects with confidence intervals, check segment effects and guardrails, assess costs and incremental value, correct for multiple testing if relevant, and recommend scale/iterate/stop with assumptions and limitations.

Executive Case-Interview Checklist

- Clarify the decision and define the target population, unit of analysis, outcome, and time window.
- Prefer randomization for causal questions where feasible; otherwise state identification assumptions and limitations.
- Pre-specify primary metrics, guardrails, sample-size logic, and stopping rules for experiments.
- Report effect sizes and confidence intervals alongside p-values; consider business value and costs.
- Segment results carefully and investigate denominator shifts, selection bias, data quality, and Simpson's paradox.
- Use time-aware validation for forecasting and compare against a sensible baseline.
- For classification, assess the confusion matrix, precision, recall, calibration, and operational threshold—not accuracy alone.
- End with a clear recommendation, confidence level, limitations, and next steps.

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STATISTICS INTERVIEW PREPARATION QUESTION BANK

LEVEL 5 — FOUNDATIONAL MACHINE LEARNING & DATA SCIENCE

60 Questions | Statistical Concepts | Solved Numericals | Selective Follow-ups

Data Analytics • Data Science • Foundational Machine Learning

How to Use This Question Bank

- Attempt each question before reading the answer. Explain the statistical reasoning and its practical implication.
- For numerical questions, show the formula, substitution, calculation, and interpretation.
- Keep data preparation inside the training workflow to prevent leakage from validation or test data.
- Difficulty stars indicate expected reasoning:
  ★☆☆ foundational,
  ★★☆ intermediate,
  ★★★ advanced.
- Follow-up questions are selective; each retained follow-up answer appears immediately below its question.

Level 5 Coverage Map

Topic	Coverage
Statistics and ML foundations	Statistical role, generalization, bias-variance, overfitting and underfitting
Data preparation	Splitting, scaling, imputation, missingness mechanisms, outliers
Model development	Cross-validation, baselines, hyperparameters, regularization
Classification statistics	Confusion matrix, accuracy, precision, recall, F1, ROC-AUC, calibration
Regression evaluation	MAE, RMSE, baselines, confidence and prediction intervals
Model reliability	Leakage, drift, feature importance, multicollinearity, bootstrap
Statistical reasoning	Hypothesis testing, causality, uncertainty, model comparison
Applied data science	Class imbalance, deployment monitoring, loan-default workflow

SECTION A — FOUNDATIONAL ML & DATA SCIENCE

Q1. What is the role of statistics in machine learning? [★ ☆ ☆]

ANSWER

Statistics helps describe data, quantify uncertainty, estimate relationships, test hypotheses, evaluate models, and distinguish signal from noise.

Q2. Differentiate descriptive statistics, statistical inference, and machine learning. [★ ☆ ☆]

ANSWER

Descriptive statistics summarize observed data; inference estimates population quantities or evaluates hypotheses under assumptions; machine learning focuses on learning patterns that generalize to new data, often for prediction.

Q3. What is the difference between a population parameter and a model parameter? [★ ★ ☆]

ANSWER

A population parameter describes a population, such as its mean. A model parameter is learned or estimated within a model, such as a regression coefficient or neural-network weight.

Q4. Why do machine-learning models need training, validation, and test data? [★ ☆ ☆]

ANSWER

Training data fit model parameters; validation data support model selection and tuning; test data provide a final estimate of generalization on unseen data.

Q5. What is overfitting, and how is it related to variance? [★ ☆ ☆]

ANSWER

Overfitting occurs when a model captures noise or idiosyncrasies in training data and performs poorly on new data. It is often associated with high variance.

FOLLOW-UP QUESTION

Can a model with excellent training performance still be unsuitable for deployment?

Answer: Yes. It may overfit or rely on leakage and perform poorly on unseen data.

Q6. Differentiate underfitting and overfitting. [★ ☆ ☆]

ANSWER

Underfitting means the model is too simple to capture important structure and often has high bias. Overfitting means it fits training-specific noise and often has high variance.

Q7. Explain the bias-variance trade-off. [★ ★ ☆]

ANSWER

Prediction error can arise from systematic model error (bias), sensitivity to the training sample (variance), and irreducible noise. Model complexity often trades bias against variance.

Q8. What is generalization in machine learning? [★ ☆ ☆]

ANSWER

Generalization is the ability of a fitted model to perform well on new observations drawn from the intended deployment setting.

Q9. Why is data leakage dangerous? [★ ☆ ☆]

ANSWER

Leakage gives the model information unavailable at prediction time or lets evaluation data influence training, causing performance estimates to look better than real deployment performance.

Q10. A dataset has 10,000 rows. You allocate 70% to training, 15% to validation, and 15% to test. How many rows go in each set? [★ ☆ ☆]

ANSWER

Multiply the total row count by each allocation.

SOLVED NUMERICAL

Training = 10,000 × 0.70 = 7,000 rows
Validation = 10,000 × 0.15 = 1,500 rows
Test = 10,000 × 0.15 = 1,500 rows
Check: 7,000 + 1,500 + 1,500 = 10,000 rows.

Q11. What is cross-validation, and why is it useful? [★ ☆ ☆]

ANSWER

Cross-validation repeats training and validation across different data folds to estimate performance more robustly and compare models without relying on one split alone.

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Q12. When is stratified splitting useful? [★ ★ ☆]

ANSWER

It helps preserve approximate class proportions across splits, especially when a classification target is imbalanced.

Q13. Why should time-series data generally be split chronologically? [★ ★ ☆]

ANSWER

Random splitting can allow future observations to inform training while earlier observations are used for testing. Chronological splits better reflect forecasting from past to future.

FOLLOW-UP QUESTION

What is the main risk of using a random split for forecasting?

Answer: It can let future observations influence training and produce an unrealistically optimistic evaluation.

Q14. What is a baseline model, and why should you use one? [★ ☆ ☆]

ANSWER

A baseline is a simple reference prediction, such as predicting the training mean for regression or the majority class for classification. It helps establish whether a more complex model adds value.

Q15. What is feature scaling, and which models often benefit from it? [★ ★ ☆]

ANSWER

Feature scaling puts variables on comparable numeric scales. Distance-based and gradient-based models, such as k-nearest neighbours, SVMs, and many linear or neural models, often benefit; tree-based models usually need it less.

16. Differentiate normalization and standardization. [★ ☆ ☆]

ANSWER

Normalization often rescales values to a fixed range, such as 0–1. Standardization typically subtracts the training mean and divides by the training standard deviation to produce z-scores.

Q17. A feature has mean 50 and standard deviation 10. Standardize an observation of 70. [★ ★ ☆]

ANSWER

Use the z-score to measure how far the observation is from the mean in standard-deviation units.

SOLVED NUMERICAL

Formula: z = (x − mean) ÷ standard deviation
Substitute: z = (70 − 50) ÷ 10
Calculate: z = 20 ÷ 10 = 2
Interpretation: The value is 2 standard deviations above the mean.

Q18. Why should a scaler be fitted only on the training data? [★ ★ ☆]

ANSWER

Using validation or test data to calculate scaling parameters leaks information. Fit the scaler on training data, then transform validation and test data with those same parameters.

FOLLOW-UP QUESTION
What data should determine the scaler's mean and standard deviation?
Answer: Use only the training partition, then apply those learned values to validation, test, and future data.

Q19. What is the purpose of imputation? [★ ☆ ☆]

ANSWER

Imputation replaces missing values using a chosen rule or model so analysis can proceed, but the method should suit the data and missingness mechanism.

Q20. Why can mean imputation distort a feature distribution? [★ ★ ☆]

ANSWER

It can reduce observed variance, create an artificial concentration at the mean, and weaken relationships between variables.

Q21. Differentiate missing completely at random, missing at random, and missing not at random. [★★★]

ANSWER

MCAR means missingness is unrelated to observed or unobserved data; MAR means it can depend on observed information; MNAR means it depends on the missing value itself or unobserved information even after conditioning on observed data.

FOLLOW-UP QUESTION

Can missingness mechanism be determined from observed data with certainty in every case?

Answer: No. Especially for MNAR, assumptions about the missing values may not be testable from observed data alone.

Q22. How can an outlier affect a model? [★★☆]

ANSWER

It may strongly influence means, variance, regression coefficients, or loss functions such as squared error. Whether to treat it depends on whether it is an error or a valid extreme case.

Q23. Why should outliers not be deleted automatically? [★☆☆]

ANSWER

An extreme value may be a genuine rare event. Validate source quality, units, and business context before correcting, transforming, retaining, or excluding it.

Q24. What is regularization in machine learning? [★☆☆]

ANSWER

Regularization discourages overly complex models by adding a penalty or constraint, often improving generalization.

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Q25. Differentiate Ridge and Lasso regression. [★★☆]

ANSWER

Ridge uses an L2 penalty and tends to shrink coefficients toward zero. Lasso uses an L1 penalty and can set some coefficients exactly to zero.

Q26. A linear model has RSS = 80 and uses a penalty $\lambda\Sigma\beta^2 = 20$. Under a simplified Ridge objective RSS + penalty, what is the objective value? [★★☆]

ANSWER

Add the residual sum of squares and the specified penalty term.

SOLVED NUMERICAL

Objective = RSS + penalty

= 80 + 20 = 100

Interpretation: Under the stated simplified objective, the penalized objective value is 100. The penalty is not itself prediction error.

FOLLOW-UP QUESTION

Does Lasso always select the scientifically correct features?

Answer: No. Selection depends on the data, scaling, penalty, and correlated predictors; it is not a guarantee of causal or scientific importance.

Q27. What is the difference between a model hyperparameter and a model parameter? [★ ☆ ☆]

ANSWER

Parameters are learned from training data, such as regression coefficients. Hyperparameters are configuration choices set outside the fitting step, such as tree depth or regularization strength.

Q28. What is the purpose of a validation set in hyperparameter tuning? [★ ☆ ☆]

ANSWER

It compares candidate settings during development. The test set should not be repeatedly used for tuning because that indirectly overfits the test set.

Q29. What is a confusion matrix? [★ ☆ ☆]

ANSWER

It tabulates predicted versus actual classes, commonly showing true positives, false positives, true negatives, and false negatives for binary classification.

Q30. A classifier produces TP = 80, FP = 20, FN = 40, and TN = 160. Calculate accuracy. [★ ★ ☆]

ANSWER

Accuracy is the number of correct predictions divided by all predictions.

SOLVED NUMERICAL

Correct predictions = TP + TN = 80 + 160 = 240
Total = 80 + 20 + 40 + 160 = 300
Accuracy = 240 ÷ 300 = 0.80 = 80%
Interpretation: The classifier correctly classifies 80% of observations in this evaluation set.

31. Using TP = 80 and FP = 20, calculate precision. [★ ★ ☆]

ANSWER

Precision measures the fraction of predicted positives that are true positives.

SOLVED NUMERICAL

Formula: Precision = TP ÷ (TP + FP)
Substitute: 80 ÷ (80 + 20) = 80 ÷ 100
Calculate: Precision = 0.80 = 80%
Interpretation: Four out of five positive predictions are correct in this sample.

Q32. Using TP = 80 and FN = 40, calculate recall. [★★☆]

ANSWER

Recall measures the fraction of actual positive cases detected by the model.

SOLVED NUMERICAL

Formula: Recall = TP ÷ (TP + FN)
Substitute: 80 ÷ (80 + 40) = 80 ÷ 120
Calculate: Recall ≈ 0.6667 = 66.67%
Interpretation: The model detects about two-thirds of the actual positive cases.

Q33. Why can accuracy be misleading for an imbalanced target? [★☆☆]

ANSWER

A model can predict the majority class most of the time and achieve high accuracy while failing to detect the minority class of interest.

Q34. Differentiate precision and recall using a fraud-detection example. [★☆☆]

ANSWER

Precision asks what share of flagged transactions are actually fraud. Recall asks what share of all fraudulent transactions the model successfully flags.

Q35. What is the F1-score? [★★☆]

ANSWER

F1 is the harmonic mean of precision and recall. It balances both metrics but does not directly include true negatives or business costs.

FOLLOW-UP QUESTION

Why is F1 not always the best metric for a business problem?

Answer: It treats precision and recall symmetrically and ignores true negatives and explicit costs; the right metric depends on the decision.

Q36. Precision is 0.80 and recall is 0.50. Calculate the F1-score. [★★☆]

ANSWER

Use the harmonic mean of precision and recall.

SOLVED NUMERICAL

Formula: F1 = 2 × (precision × recall) ÷ (precision + recall)
Substitute: F1 = 2 × (0.80 × 0.50) ÷ (0.80 + 0.50)
Calculate: F1 = 0.80 ÷ 1.30 ≈ 0.6154
Interpretation: F1 is approximately 0.615, or 61.54%.

Q37. What is the difference between ROC-AUC and classification accuracy? [★★☆]

ANSWER

Accuracy evaluates predictions at one chosen threshold. ROC-AUC evaluates ranking discrimination across thresholds using true-positive and false-positive rates.

Q38. Why may precision-recall curves be more informative than ROC curves for rare positive classes? [★★☆]

ANSWER

Precision-recall curves focus on positive predictions and positive-case detection, making trade-offs clearer when positives are rare.

Q39. How does the classification threshold affect false positives and false negatives? [★★☆]

ANSWER

Lowering the threshold typically flags more cases, increasing recall while potentially increasing false positives. Raising it usually reduces flags but can miss more actual positives.

FOLLOW-UP QUESTION

Does a 0.70 predicted probability guarantee the event occurs?

Answer: No. It is a probability estimate across comparable cases, not a certainty for one individual case.

Q40. A model predicts a 0.70 fraud probability for a transaction. What does that probability mean, and what does it not guarantee? [★★★]

ANSWER

If well calibrated, cases assigned probabilities near 0.70 should be fraudulent about 70% of the time over comparable cases. It does not guarantee this individual transaction is fraud.

Q41. What is probability calibration? [★★☆]

ANSWER

Calibration means predicted probabilities correspond to observed event frequencies across comparable predictions. It is distinct from ranking or discrimination.

Q42. Differentiate regression and classification. [★☆☆]

ANSWER

Regression predicts a numerical outcome, such as sales. Classification predicts a category or class probability, such as churn versus no churn.

Q43. What is the difference between MAE and RMSE? [★☆☆]

ANSWER

MAE averages absolute errors and is easier to interpret in outcome units. RMSE squares errors before averaging, so larger errors have more influence.

Q44. Actual values are [10, 20, 30] and predictions are [12, 18, 33]. Calculate MAE. [★★☆]

ANSWER

Compute absolute errors and average them.

SOLVED NUMERICAL

Absolute errors: |10−12| = 2; |20−18| = 2; |30−33| = 3
MAE = (2 + 2 + 3) ÷ 3 = 7 ÷ 3 ≈ 2.33
Interpretation: Predictions differ from actual values by about 2.33 units on average in absolute terms.

Q45. Why should regression metrics be compared against a baseline? [★☆☆]

ANSWER

A baseline establishes a simple benchmark. A complex model is useful only if its performance improvement is reliable and meaningful for the task.

FOLLOW-UP QUESTION

Can a highly important feature be a poor choice for a causal intervention?

Answer: Yes. Predictive importance does not show that intervening on the feature will change the outcome.

6. What is concept drift or data drift in a deployed model? [★★☆]

ANSWER

Data drift is a change in input-data distribution. Concept drift is a change in the relationship between inputs and outcomes. Both can reduce deployed performance and should be monitored.

Q47. How can you monitor a model after deployment? [★★☆]

ANSWER

Track input quality and distributions, prediction distributions, outcome-based metrics when labels arrive, segment performance, drift signals, and operational/business guardrails.

Q48. Why is feature importance not the same as causality? [★★☆]

ANSWER

Feature importance describes how a model uses variables for prediction under a particular method. It does not establish that changing a feature will cause the outcome to change.

Q49. What is the difference between correlation and a useful predictive feature? [★★☆]

ANSWER

Correlation is a measure of association. A predictive feature must add generalizable information at prediction time; leakage, instability, or redundant information can make a correlated feature unsuitable.

Q50. How can multicollinearity affect a linear model used for interpretation? [★★☆]

ANSWER

Correlated predictors can make individual coefficient estimates unstable and increase their standard errors, even when overall

predictions remain reasonable.

Q51. What does a confidence interval for a regression coefficient contribute to model interpretation? [★★☆]

ANSWER

It communicates uncertainty around the estimated conditional association under the model assumptions, rather than reporting only a single coefficient value.

Q52. Why does a statistical association found in observational data not automatically imply causation? [★☆☆]

ANSWER

Confounding, selection effects, reverse causality, and measurement issues can explain observed associations. Causal claims need an appropriate design and assumptions.

FOLLOW-UP QUESTION

Why must oversampling happen inside each training fold?

Answer: Oversampling before splitting can leak duplicated or related examples into validation/test data and inflate measured performance.

Q53. How would you use a hypothesis test when evaluating a model improvement? [★★★]

ANSWER

Define the comparison and metric in advance, use an evaluation design that respects data dependence, estimate uncertainty in the difference, and consider practical value alongside statistical evidence.

Q54. What is a bootstrap, and how can it help model evaluation? [★★☆]

ANSWER

Bootstrap resampling repeatedly samples observations with replacement to estimate uncertainty in statistics or metrics, provided the resampling scheme matches the data structure.

Q55. Why can random bootstrap resampling be inappropriate for time-dependent data? [★★★]

ANSWER

It breaks temporal dependence and ordering. Use a time-series-aware resampling or evaluation method appropriate to the forecasting problem.

Q56. What is the difference between a confidence interval and a prediction interval? [★★☆]

ANSWER

A confidence interval describes uncertainty in an estimated population quantity or mean response. A prediction interval describes a future individual outcome and is usually wider because it includes outcome noise.

Q57. How can class imbalance be addressed, and what should be checked? [★★★]

ANSWER

Options include class weighting, resampling, threshold adjustment, or collecting more representative data. Evaluate precision, recall, calibration, and performance on a realistic holdout; resampling must occur only inside training folds.

FOLLOW-UP QUESTION

What is one advantage of bootstrap intervals for a complex metric?

Answer: They can estimate uncertainty for statistics whose sampling distribution is difficult to derive analytically, if resampling assumptions are appropriate.

Q58. What is the role of statistical significance in ML model comparison? [★★★]

ANSWER

It can help assess uncertainty in a performance difference, but the evaluation design must account for dependence and repeated comparisons. Practical importance and generalization also matter.

Q59. What should you do if a complex model performs only marginally better than a simple model? [★★☆]

ANSWER

Compare uncertainty, maintenance cost, interpretability, latency, fairness, and robustness. Prefer the simpler model when the small gain does not justify the added complexity.

Q60. Final case: a bank wants to predict loan default. Outline a statistically sound ML workflow. [★★★]

ANSWER

Define the target and decision point; inspect data quality and leakage; create time-appropriate train/validation/test splits; handle missingness and class imbalance within training; establish a baseline; tune on validation data; evaluate discrimination, calibration, threshold costs, and subgroup performance; monitor drift; and document uncertainty, limitations, and governance.

ML & Data Science Interview Checklist

- Clarify the prediction target, unit of analysis, prediction time, and intended decision.
- Create splits that match deployment; use chronological splits for time-dependent data.
- Fit preprocessing, feature selection, and resampling on training data only.
- Compare models with a baseline and choose metrics aligned with business costs.
- Interpret classification metrics together and consider probability calibration.
- Distinguish prediction from causation and feature importance from intervention effects.
- Monitor data quality, drift, subgroup performance, and operational outcomes after deployment.
- Report uncertainty, assumptions, limitations, and the decision supported by the model.